Sectigo CLM and ADCS Integration Guide

Step-by-step integration of Sectigo Certificate Manager (SCM) with a Microsoft AD CS issuing CA through a CA connector, with a test plan and troubleshooting.

Stage 1: Prerequisites

1. What you are building

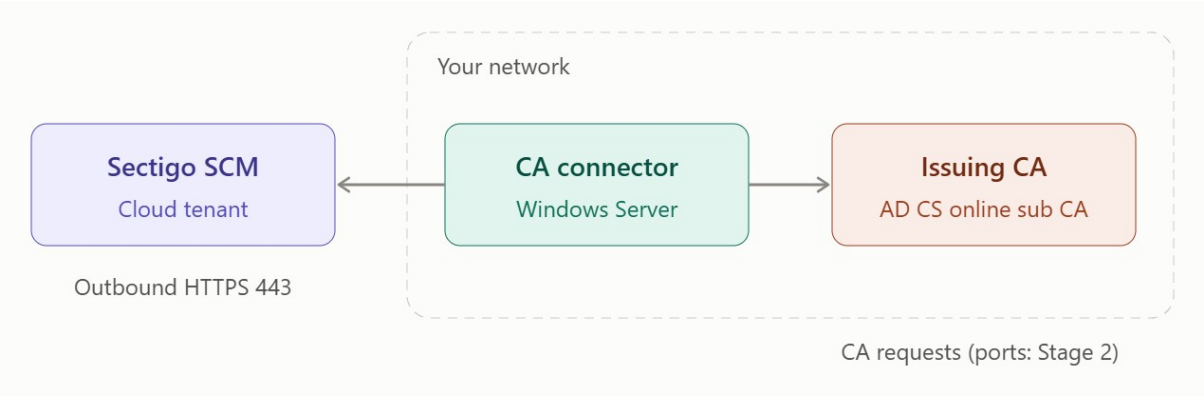


Figure 1. *Architecture: Sectigo SCM, CA connector and issuing CA*

A **CA connector** is an agent that runs on a Windows server inside your network and links your Microsoft CA to Sectigo's cloud. Sectigo CA connectors integrate with SCM to manage certificates issued by Microsoft CA, along with AWS Private CA, DigiCert, Entrust and Google CAS. Your CA is never exposed to the internet, and the connector only talks to your online issuing CA. The offline root stays offline.

Inside SCM, the integration is built as a chain: **CA connector → CA backend → certificate profile**. The connector must be linked to a CA backend in SCM, and that backend is attached to the profiles used for issuance. Stages 2 and 3 build this chain.

For Microsoft CA, SCM supports private SSL, client, device and code signing certificates. It does not support public DV, OV or EV SSL on this path.

2. Sectigo-side prerequisites

#	Item	Why it matters
S1	Trial tenant is active and you can log in as an administrator	Needed to add the connector and download its installer
S2	Know your tenant URL: `cert-manager.com`, `hard.cert-manager.com` or `eu.cert-manager.com`	The connector docs list all three as endpoints, and the one matching your tenant goes into the firewall rules. For a European client, eu is likely. Check the browser address bar.
S3	Confirm your trial includes the Microsoft CA connector	Trials can have features disabled. Ask your Sectigo contact before investing time.

3. Connector host prerequisites

- 64-bit Windows Server 2019 or 2022, with local admin rights to install.
- A dedicated, domain-joined member server in the same network zone as the CA. A small VM is enough (suggested for a trial: 2 vCPU, 4 GB RAM, 40 GB disk).
- Default install path: `C:\Program Files\Sectigo Limited\SectigoCBCS`. Plan Defender exclusions and monitoring for that path.
- Correct time sync. Kerberos and TLS both break with clock skew.

> **Note:** The Sectigo documentation does not say whether the connector can run on the issuing CA itself. A separate server is recommended.

4. AD CS prerequisites

#	Requirement	Why
A1	Online issuing CA healthy, valid CA certificate, fresh CRL	The connector can only issue from a working CA
A2	The default Enrollment Agent (Computer) template published on the CA (do not duplicate it)	The connector enrolls on behalf of others
A3	Every template SCM uses must grant Read and Enroll to the CA connector	Otherwise issuance fails with access denied
A4	Only if the subject is built from AD data: the template needs one authorized signature and the Certificate Request Agent application policy under Issuance Requirements	Required by the "Build Subject from AD information" option
A5	A lab or non-production issuing CA	Keeps the trial away from production
A6	Duplicated test templates, not live ones	Prevents any effect on real enrollment

Health checks to run on the CA now (replace the names):

```
certutil -config "CAHOST\CA-Name" -ping
certutil -config "CAHOST\CA-Name" -CATemplates
certutil -config "CAHOST\CA-Name" -CAInfo
```

5. Network prerequisites

- **Connector host outbound:** HTTPS 443 to your SCM tenant. The exact endpoints are listed in Stage 2. Check whether a proxy or TLS inspection sits in the path.
- **Connector host to issuing CA:** the ports used for remote requests to a Microsoft CA. This is usually RPC (135 plus the dynamic range), but the Sectigo documentation does not state it, so confirm it in Stage 2 before changing firewalls.
- **DNS** for both the SCM endpoint and the CA.

6. Accounts and permissions

Where	Account	Needs
SCM	Your trial admin	Rights to add connectors, CA backends and certificate profiles (exact role verified in Stage 3)

Where	Account	Needs
Connector host	Your admin	Local admin to install
AD CS	You or a PKI admin	CA Manager rights, plus write access on templates to set permissions
AD	The connector's identity (the computer account, see Stage 2)	Read and Enroll on test templates, Enroll on the Enrollment Agent template

Stage 2: Infra readiness

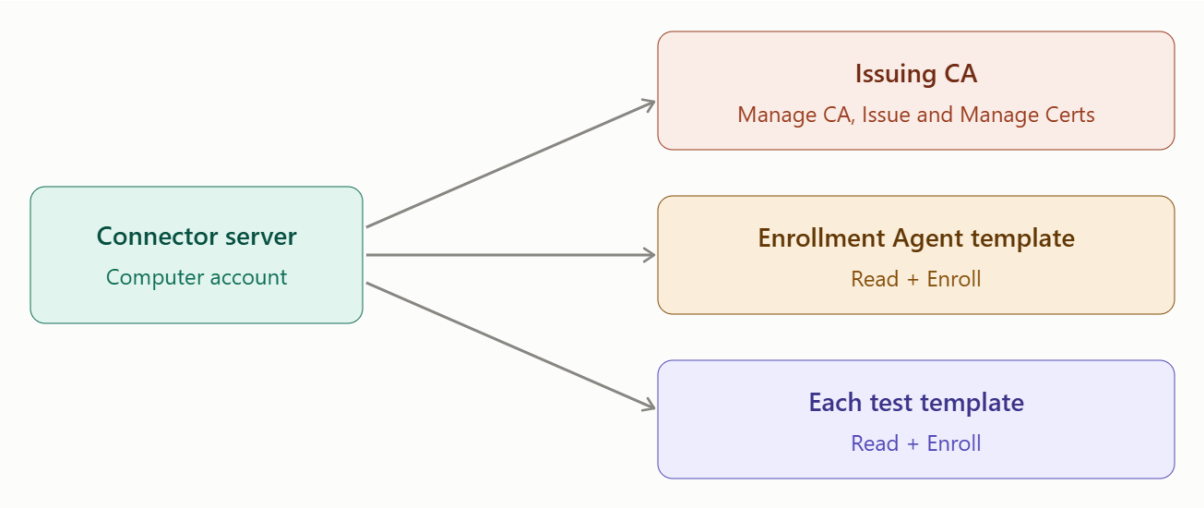


Figure 2. Permissions the connector's computer account needs

The connector's identity is the **computer account** of the machine it is installed on (`QS-CA-SCM$`), as the diagram shows. That account needs rights on the CA, the Enrollment Agent template and each test template.

Outbound firewall rules. The connector needs outbound TCP 443 to `dist.sectigo.com` and to the SCM instance for your tenant:

- `cbcc.enterprise.sectigo.com` (main instance; used when the tenant is on `cert-manager.com`)
- `cbcc.hard.sectigo.com`
- `cbcc.eu.sectigo.com`

The connector install page lists these endpoints. Confirm your instance with Sectigo if unsure. This guide uses `cbcc.enterprise.sectigo.com`.

Step 1: Prepare the connector server

1. Build a Windows Server 2019 or 2022 (64-bit) VM and join it to the domain. Sectigo's stated minimum is 2 GB RAM and a 1.4 GHz CPU.
2. Patch it, set the hostname, and check time sync:

```
w32tm /query /status
```

3. Plan Defender exclusions for the install path and for `C:\ProgramData\Sectigo Limited\SectigoCBCS`, where the connector keeps its config and logs.
4. Note your proxy details, if any. The installer offers direct or manual proxy settings, with a Test Connection button.

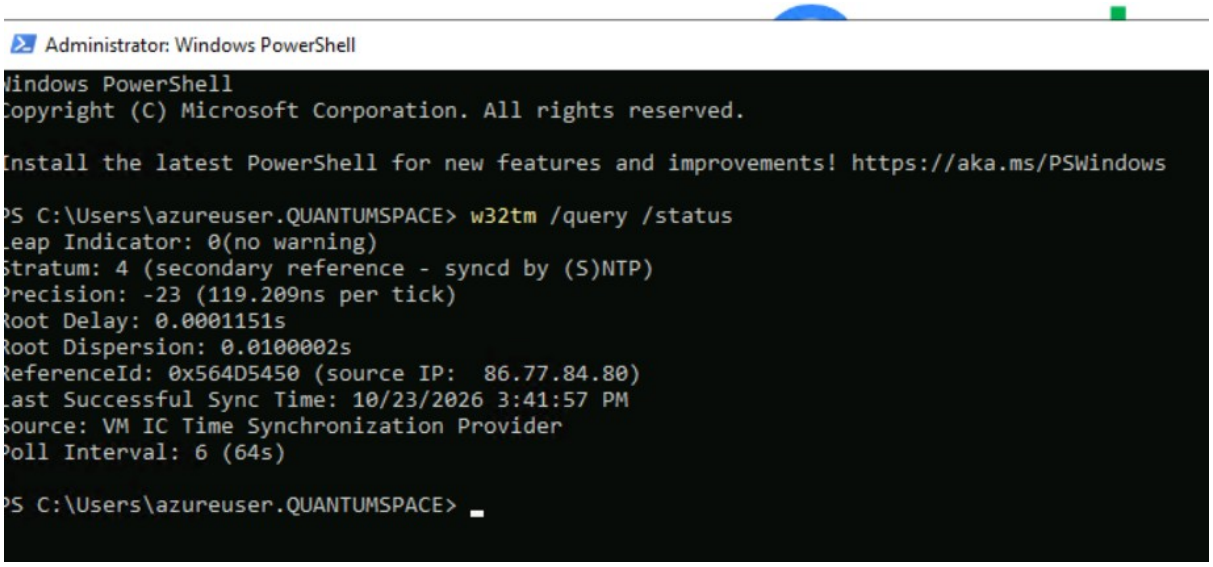


Figure 3. Time sync status on the connector server

Step 2: Open the network paths

Outbound to Sectigo from `QS-CA-SCM`, TCP 443: `dist.sectigo.com` and `cbcc.enterprise.sectigo.com`. Test:

```
Test-NetConnection dist.sectigo.com -Port 443
Test-NetConnection cbcc.enterprise.sectigo.com -Port 443
```

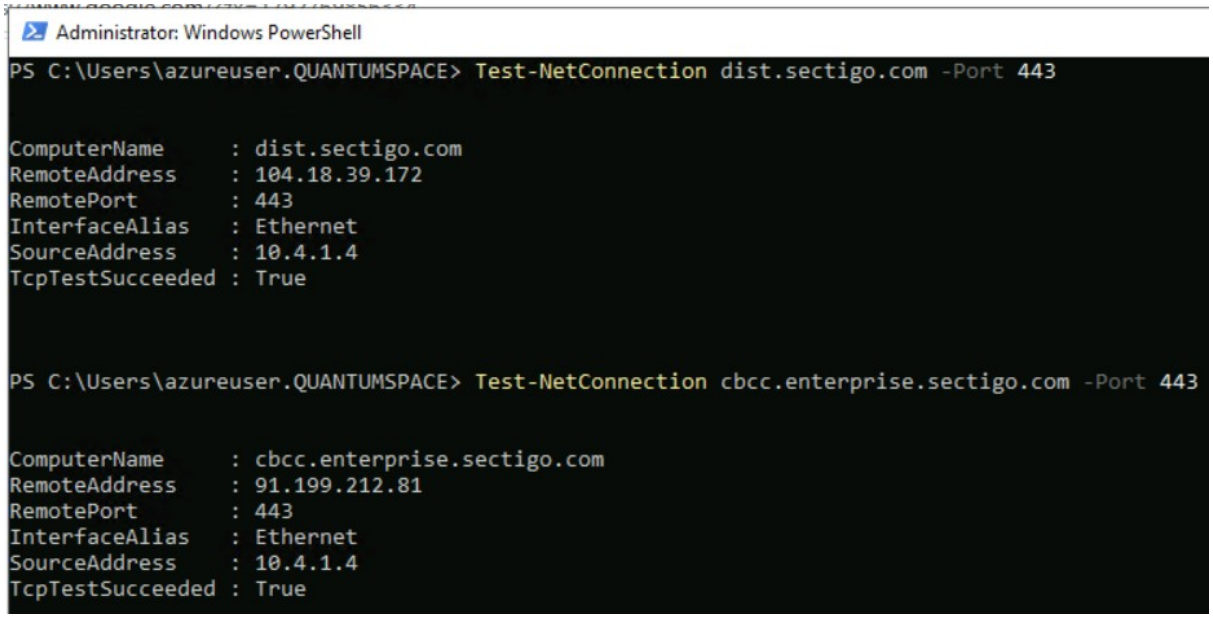


Figure 4. Outbound port 443 test to the Sectigo endpoints

If you use TLS inspection, exempt these hosts.

`QS-CA-SCM` to the issuing CA: normal remote Microsoft CA requests use RPC/DCOM, so check TCP 135 plus the dynamic RPC range:

```
Test-NetConnection QS-SUB-CA1 -Port 135
certutil -config "QS-SUB-CA1\Quantumspace Issuing CA1" -ping
```

If `-ping` works from `QS-CA-SCM`, the path is good.

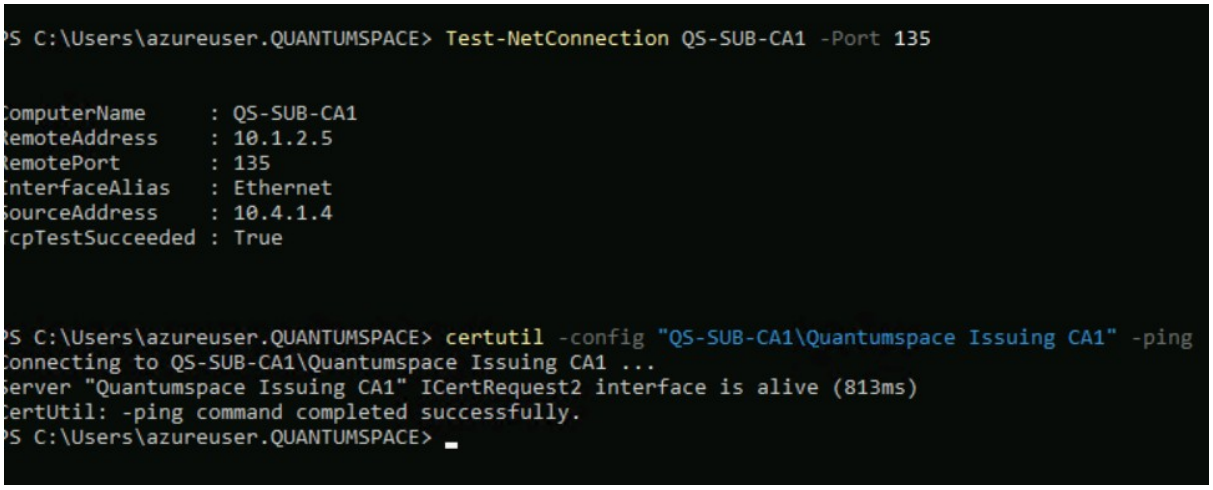


Figure 5. *RPC port 135 test and CA ping from the connector server*

Step 3: Publish the Enrollment Agent template

The CA needs the **default** Enrollment Agent (Computer) template published, with the connector machine granted Read and Enroll. Do not duplicate the template.

1. On a machine with the CA tools, open `certtmpl.msc`.
2. **Do not duplicate** Enrollment Agent (Computer). Set the permissions on the default template itself: right-click it → Properties.
3. On the Security tab, click Add, set Object Types to include Computers, add `QS-CA-SCM$`, and tick **Read** and **Enroll**.
4. Open `certsrv.msc` on the issuing CA, right-click Certificate Templates, then **New → Certificate Template to Issue**, and pick Enrollment Agent (Computer).
5. Check the CA's Enrollment Agents tab. If restrictions are enabled, `QS-CA-SCM$` must be allowed for your test templates. This is general Microsoft CA behavior, not from the Sectigo documentation.

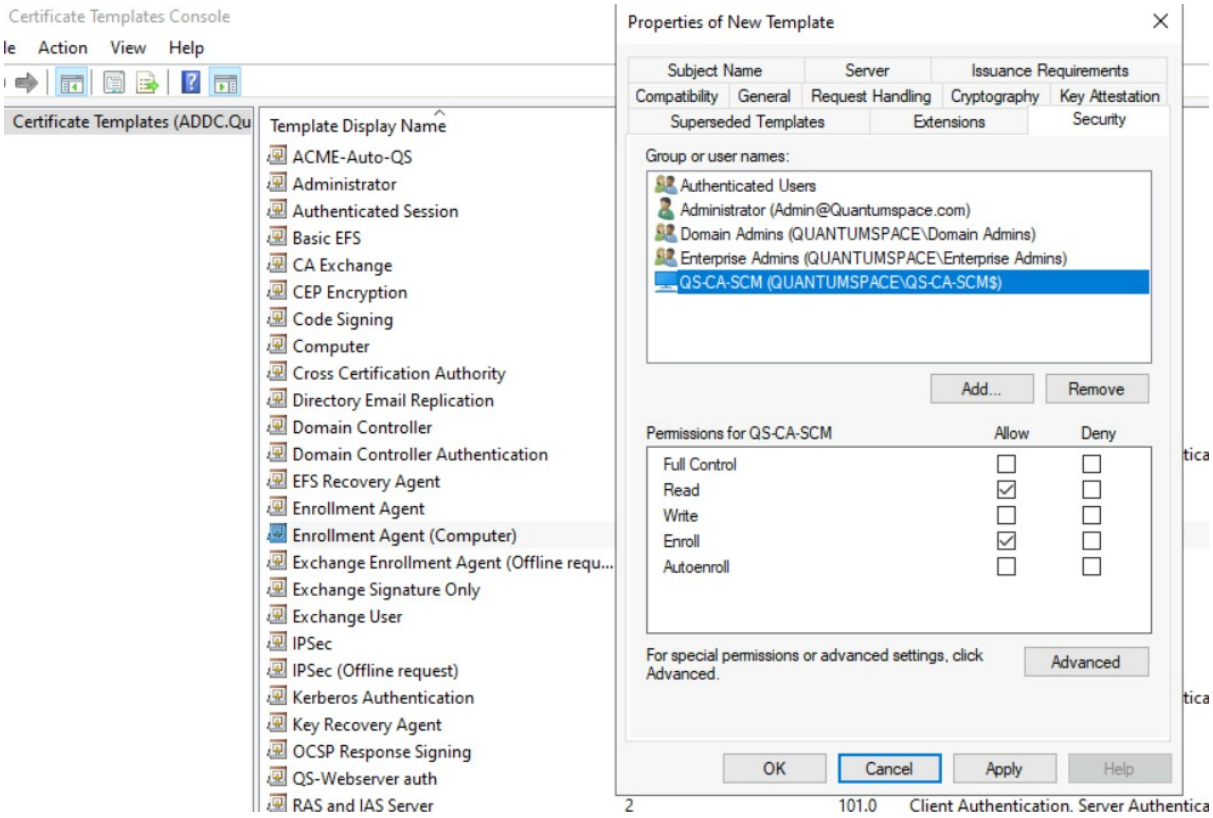


Figure 6. Security tab of the Enrollment Agent template

Step 4: Give the connector rights on the CA

The connector machine must hold both **Manage CA** and **Issue and Manage Certificates** on the issuing CA.

1. In `certsrv.msc`, right-click the CA → Properties → Security.
2. Add `QS-CA-SCM$` and allow both permissions.

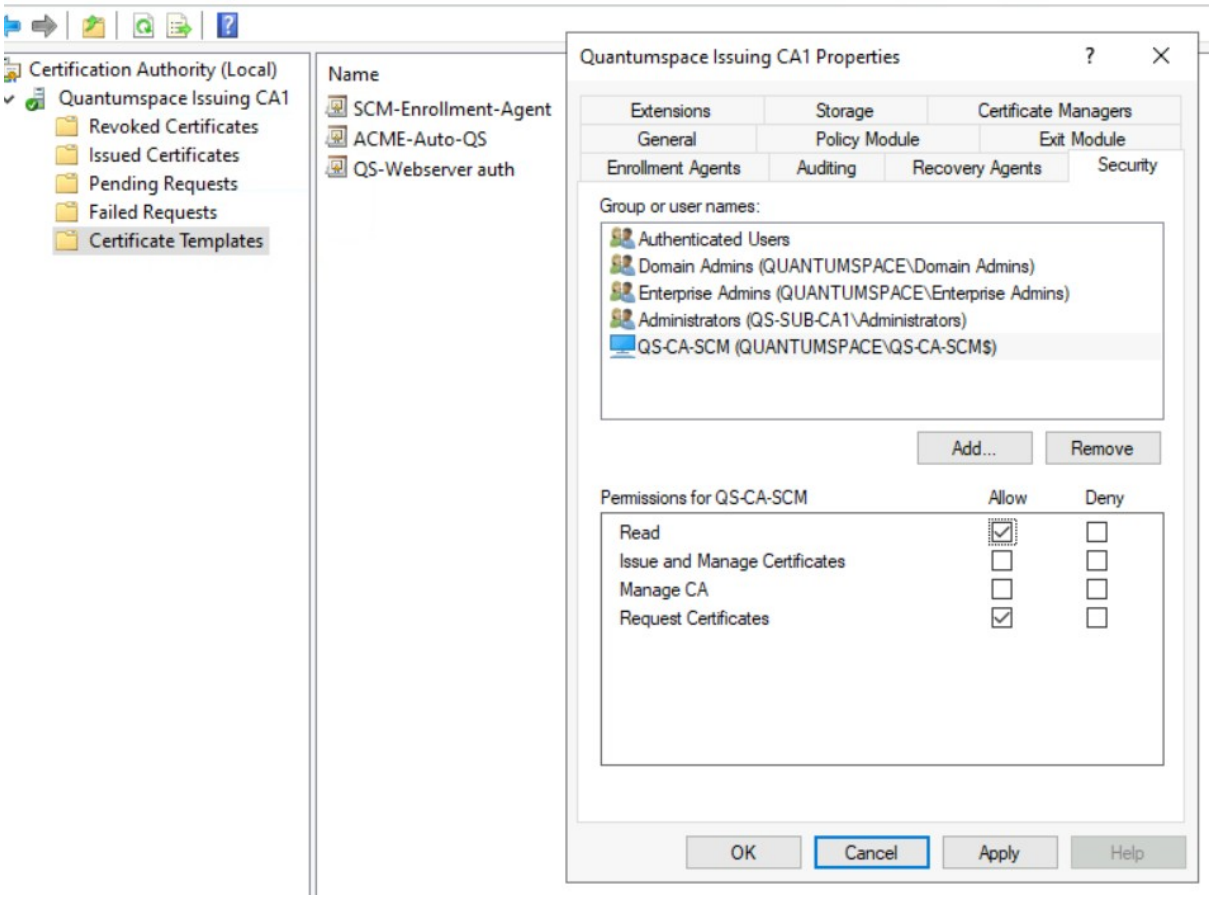


Figure 7. Manage CA and Issue and Manage Certificates on the issuing CA

Step 5: Prepare test templates

1. Duplicate a simple client template and a web server template. Name them `SCM-Test-Client` and `SCM-Test-WebServer`.
2. On each, add `QS-CA-SCM$` with Read and Enroll. All Microsoft templates used must grant the connector Read and Enroll access.
3. Set the subject to **Supply in the request** for the first test. This is the simplest path.
4. To build subjects from AD later, the template needs one authorized signature and the Certificate Request Agent application policy. Leave that for Stage 4 or later.
5. Publish both templates on the issuing CA.

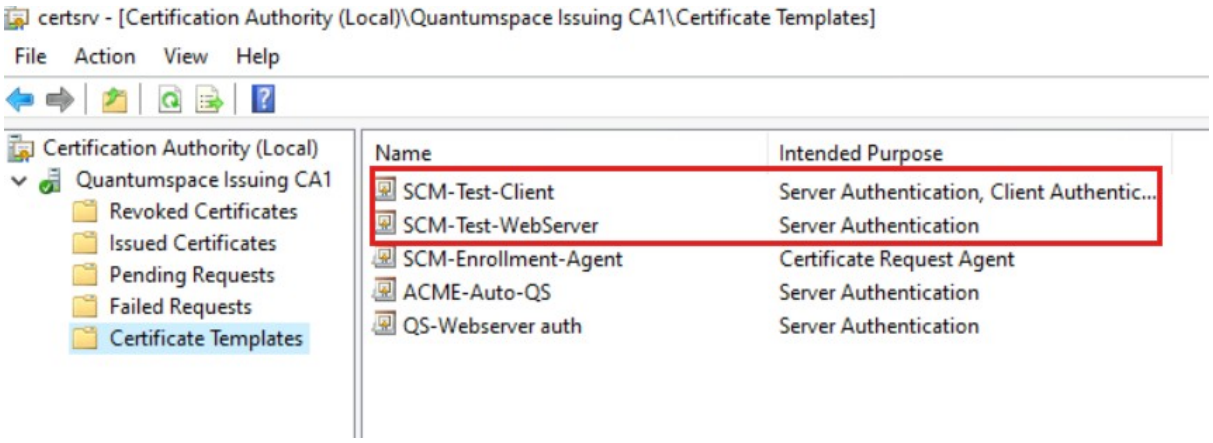


Figure 8. Test templates published on the issuing CA

Step 6: Check your SCM rights

Installing the connector needs an SCM account with **MRAO** permissions and administrator permissions on the CA. In your tenant, open your profile or the Administrators page and confirm your role is MRAO.

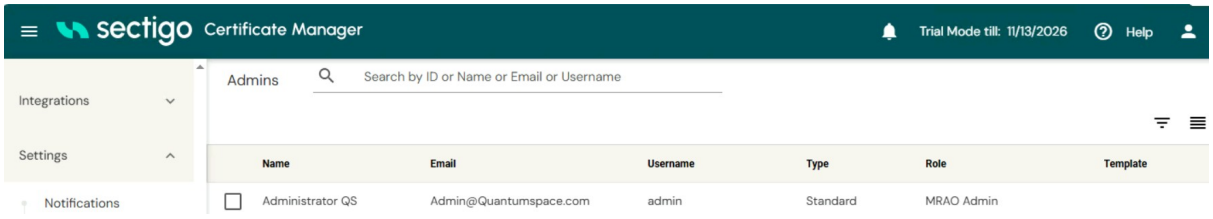


Figure 9. SCM administrator account with the MRAO role

Stage 3: Implementation

The diagram shows the order. Steps 1, 4 and 5 happen in the SCM console. Steps 2 and 3 happen on `QS-CA-SCM`.



Figure 10. Stage 3: order of work

Step 1: Add the connector in SCM

1. Go to **Integrations → CA Connectors** and click the Add icon.
2. Enter a name and optional comments, then click Next.
3. Copy the installation token. If an install fails, the next attempt needs a new token.
4. Download the Windows package and click Save.

The connector now appears with status **Pending**.

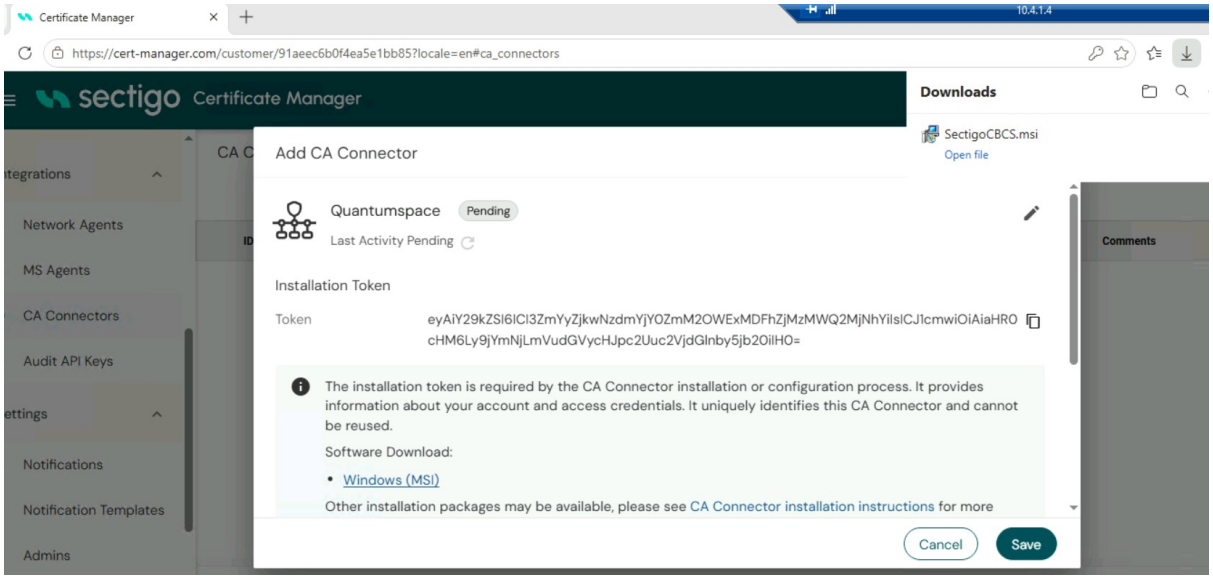


Figure 11. *Add CA Connector in SCM, with the installation token*

Step 2: Install it on QS-CA-SCM

Copy `SectigoCBCS.msi` to `QS-CA-SCM`, then use either method.

GUI

1. Right-click the MSI → Install → Next → accept the EULA → Next.
2. Keep the default path (`C:\Program Files\Sectigo Limited\`) → Next.
3. Paste the token → Next.
4. On Proxy Settings pick Direct or enter your proxy, and click Test Connection.
5. Install → Yes at the prompt → Finish.

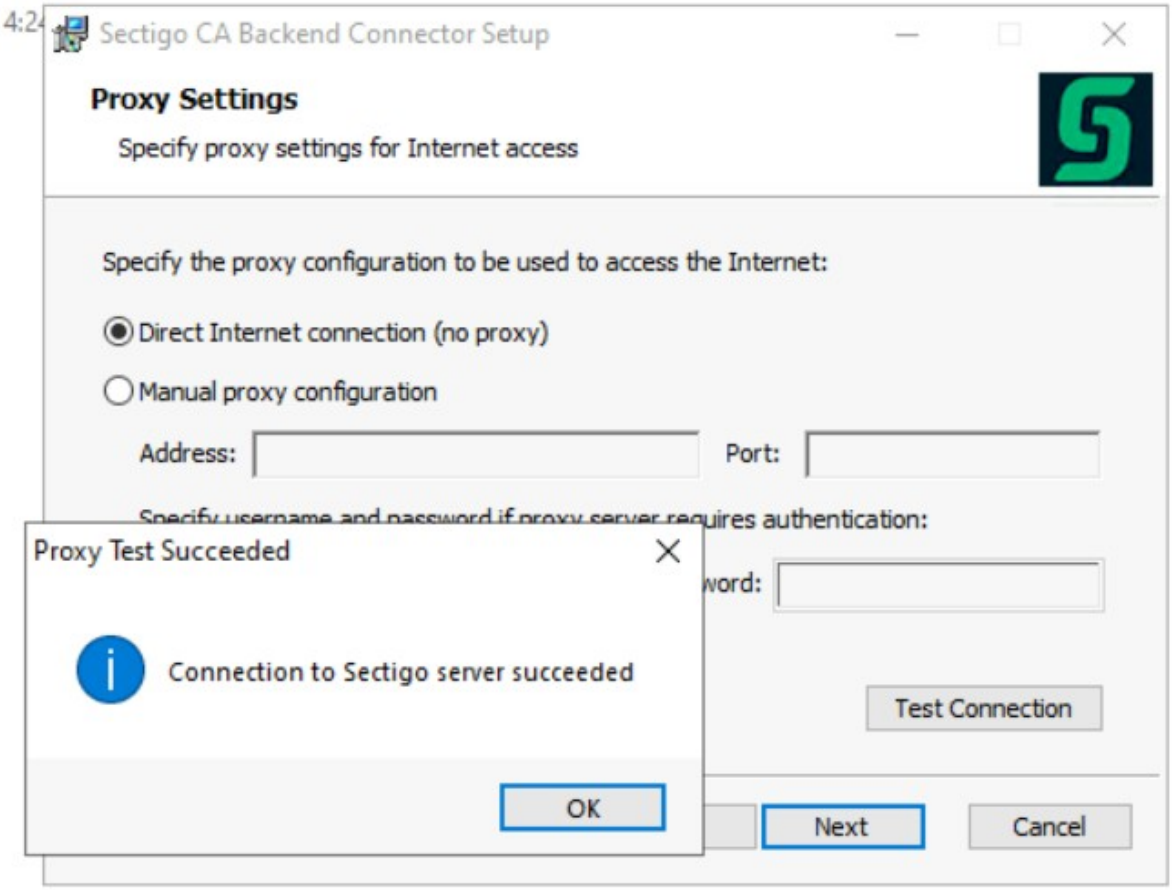


Figure 12. Connector installer: proxy settings and connection test

Silent CLI (run from the folder containing the MSI):

```
msiexec.exe /i SectigoCBCS.msi /q TOKEN=<your_token>
```

The Sectigo documentation spells the file name two ways. Use whatever name your downloaded file has, and drop unused proxy options.

Verify: in SCM the status should change to **Connected**. Config and logs live in `C:\ProgramData\Sectigo Limited\`. If it stays Pending, check the 443 paths from Stage 2 and read those logs.

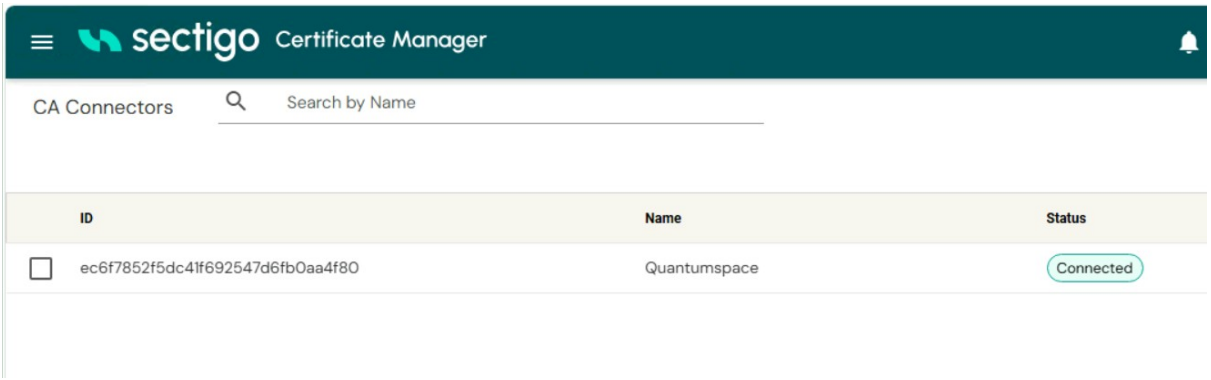


Figure 13. Connector shown as Connected in SCM

> **Recommended:** enable the **CA Connector Disconnected** notification so you are alerted if the connector drops.

Step 3: Create the local backend on QS-CA-SCM

Installing the connector is not enough. The connector must also be configured to reach your CA, using the `sectigo-cbcs.exe` tool.

1. Open an elevated Command Prompt on `QS-CA-SCM` and go to the install folder (`C:\Program Files\Sectigo Limited\`).
2. Create the backend:

```
sectigo-cbcs.exe backend add -name MSCA-LAB -type msca -server QS-SUB-CA1 -ca "Quantumspace Issuing CA1"
```

3. Enroll the agent certificate:

```
sectigo-cbcs.exe backend msca enroll-agent-cert -name MSCA-LAB -ca "Quantumspace Issuing CA1"
```

4. If a command fails, `sectigo-cbcs help` lists the options.

The documentation gives an example with the type `msca`, the CA server's hostname and the CA's common name. Get the common name with `certutil -config "CAHOST\CA-Name" -CAInfo name`.

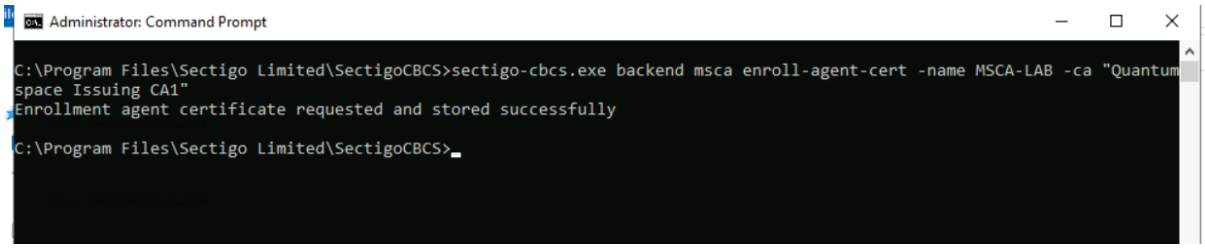


Figure 14. *Enrollment agent certificate requested and stored*

The enrollment agent certificate is issued to the connector machine with the Certificate Request Agent purpose.

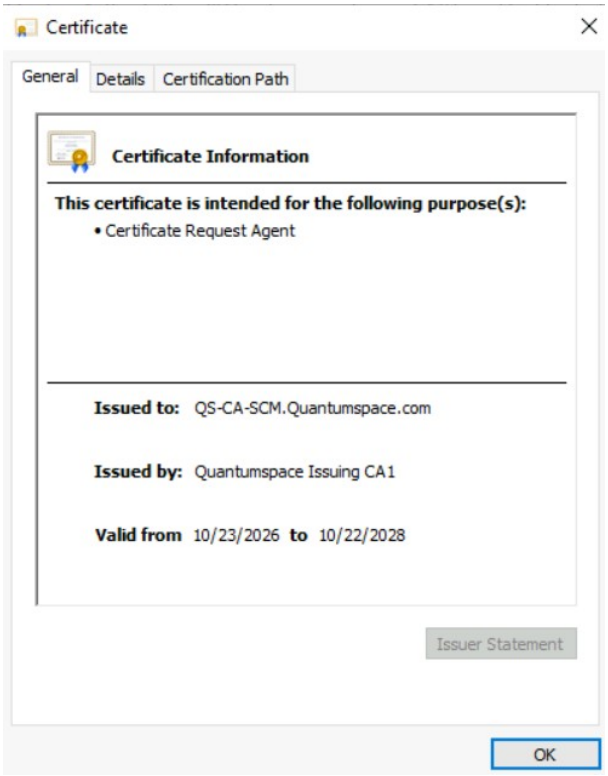


Figure 15. *Enrollment agent certificate (Certificate Request Agent purpose)*

> **Remember the name `MSCA-LAB`.** SCM asks for it in the next step.

Step 4: Create the CA backend in SCM

1. Go to **Issuers → CA Backends** and click Add.
2. Fill in the fields below.
3. Click Save.

- **Backend Type:** Microsoft
- **Name:** Q-space
- **Connector:** Quantumspace
- **Local CA Backend:** MSCA-LAB (the name given when creating the local backend)

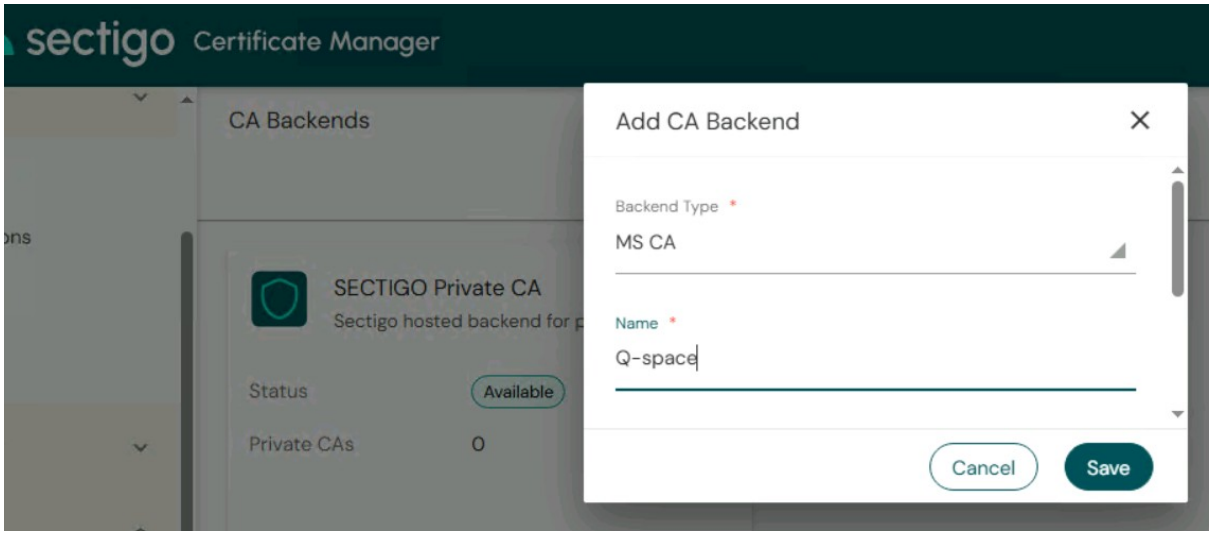


Figure 16. *Add CA Backend in SCM: type and name*

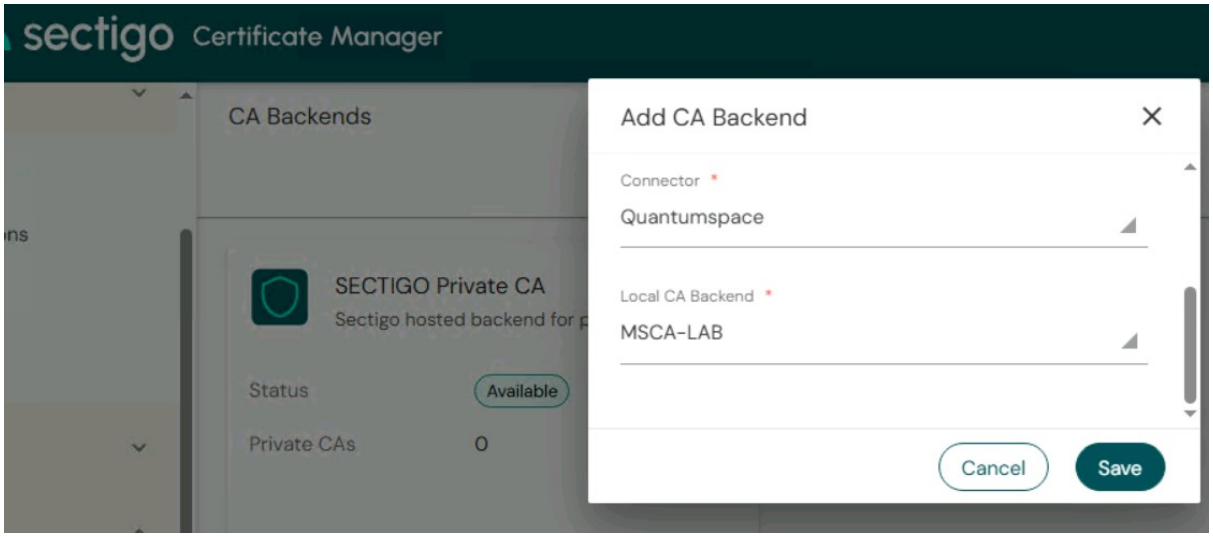


Figure 17. *Add CA Backend in SCM: connector and local backend*

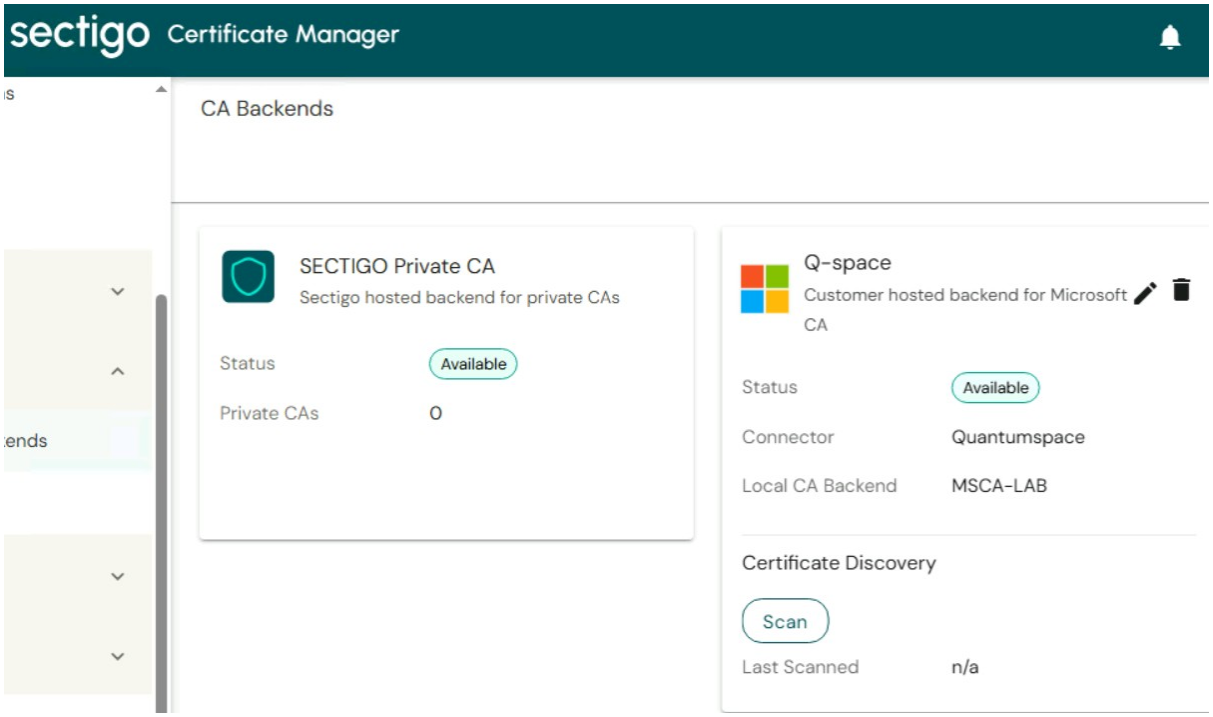


Figure 18. CA backend listed in SCM

Step 5: Create certificate profiles

A profile is what users or automation request certificates against. Make two, under **Enrollment → Certificate Profiles → Add**.

Page 1

- **Name:** `QS-Test-WebServer`
- **CA Backend:** Q-space
- **Certificate Type:** SSL

Page 2 (Microsoft)

- **Issuing CA:** the CA's Common Name
- **MS Template:** `SCM-Test-WebServer`. The connector needs Read and Enroll on it, which you already set.
- **Terms:** validity period (keep it short for the lab, such as 40 days)
- **Allowed Key Types:** for example RSA 2048, plus ECC if your template allows it
- **Requires approval:** off for now, so tests flow without a second admin
- **Allow Renew:** on
- **Domain Policies:** SSL profiles have this field. Start permissive for the lab and tighten it in operations.
- **Build Subject from AD information:** leave off for now

Repeat for `QS-Test-Client` (type Client, MS Template `SCM-Test-Client`). Client profiles also have **Auto Revoke** and **Max Number of Valid Certificates**. Leave both at defaults for the first test.

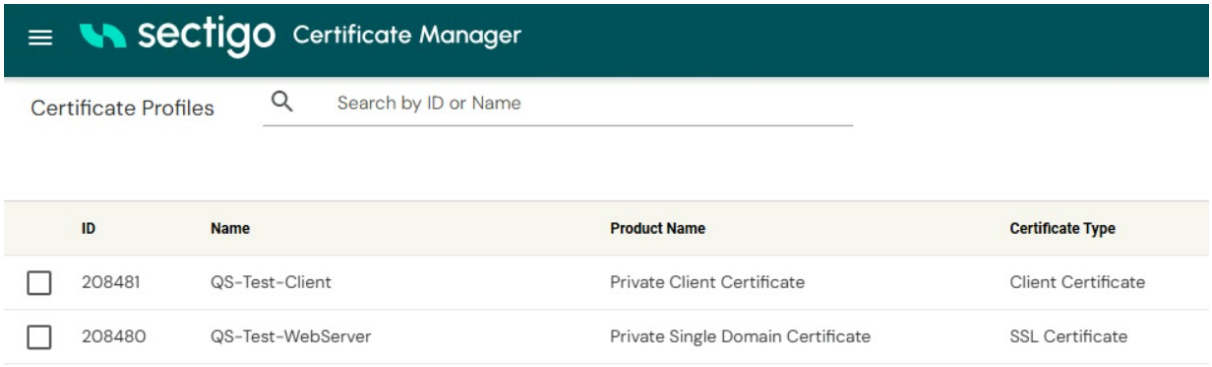


Figure 19. Certificate profiles in SCM

Stage 4: Testing

The diagram shows what each test exercises. Any failure tells you which hop broke: SCM, the connector, or the CA.

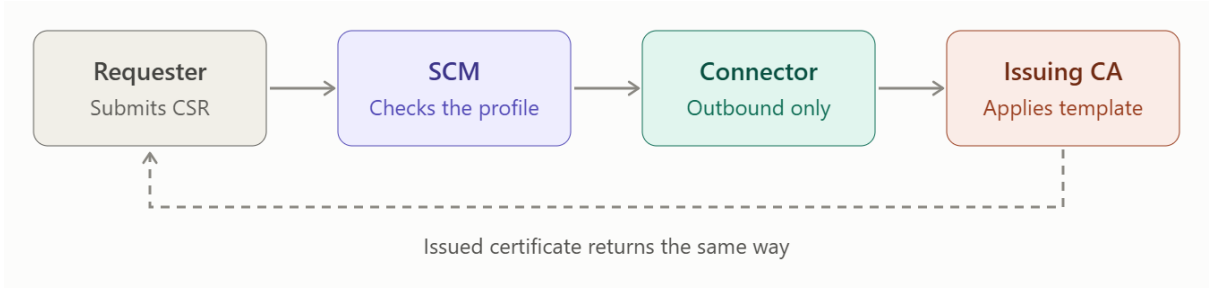


Figure 20. How a request flows through SCM, the connector and the CA

Test 0: Pre-flight (2 minutes)

- Connector status is **Connected** (Integrations → CA Connectors).
- `certutil -config "QS-SUB-CA1\Quantumspace Issuing CA1" -CATemplates` lists `SCM-Test-WebServer` and `SCM-Test-Client`.
- Both profiles exist under Enrollment → Certificate Profiles.

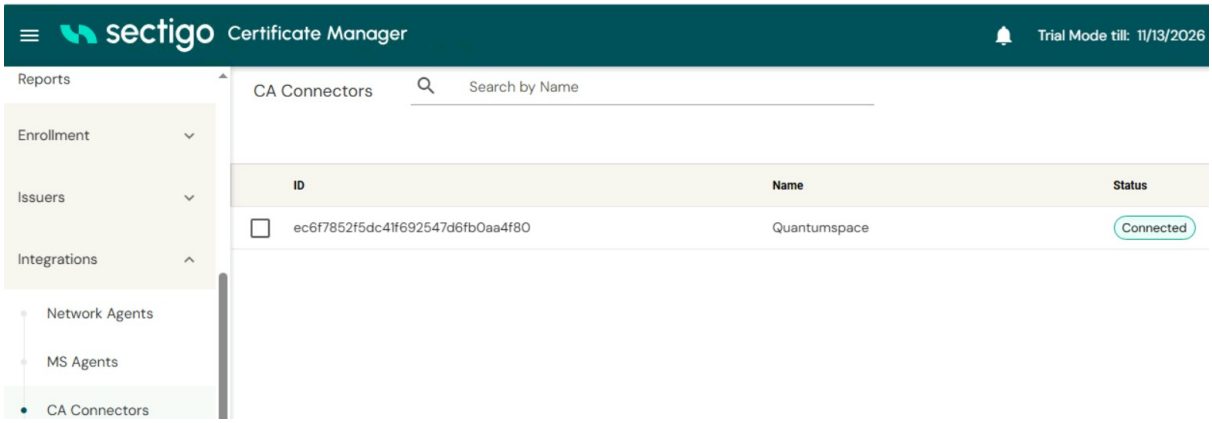


Figure 21. Connector status in SCM

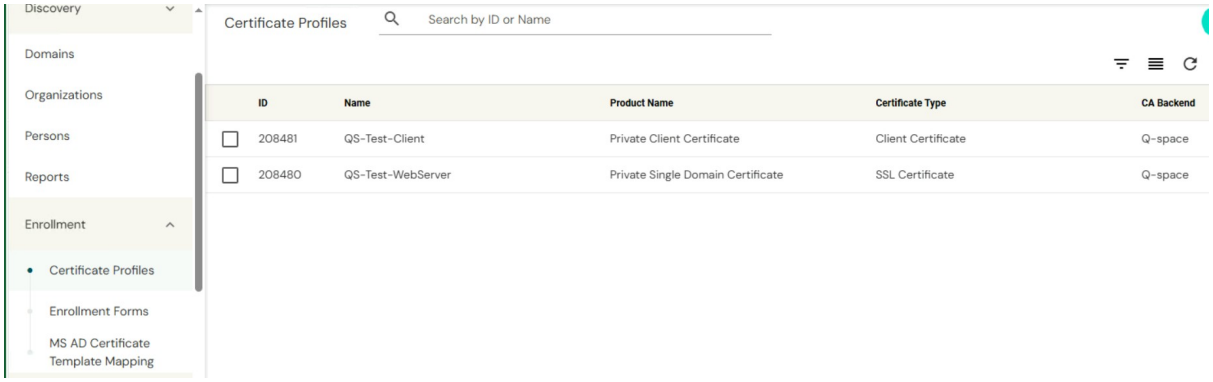


Figure 22. *Certificate profiles in SCM*

> **Note:** Profiles are delegated to an organization. Sectigo's walkthrough for another profile type includes a step to assign the profile to the organization. This has not been confirmed for private SSL profiles, so if a profile is missing at enrollment, check your organization's profile tab.

Test 1: Web server certificate (the main test)

1a. Create a CSR on any Windows machine

Save as `scmtest.inf` (replace the hostname with a real name in your domain):

```
[Version]
Signature="$Windows NT$"

[NewRequest]
Subject = "CN=QS-CA-SCM.Quantumspace.com"
KeySpec = 1
KeyLength = 3072
Exportable = FALSE
MachineKeySet = TRUE
ProviderName = "Microsoft RSA SChannel Cryptographic Provider"
ProviderType = 12
RequestType = PKCS10
KeyUsage = 0xa0

[Extensions]
2.5.29.17 = "{text}"
_continue_ = "dns=QS-CA-SCM.Quantumspace.com"
```

Then:

```
certreq -new scmtest.inf scmtest.csr
```

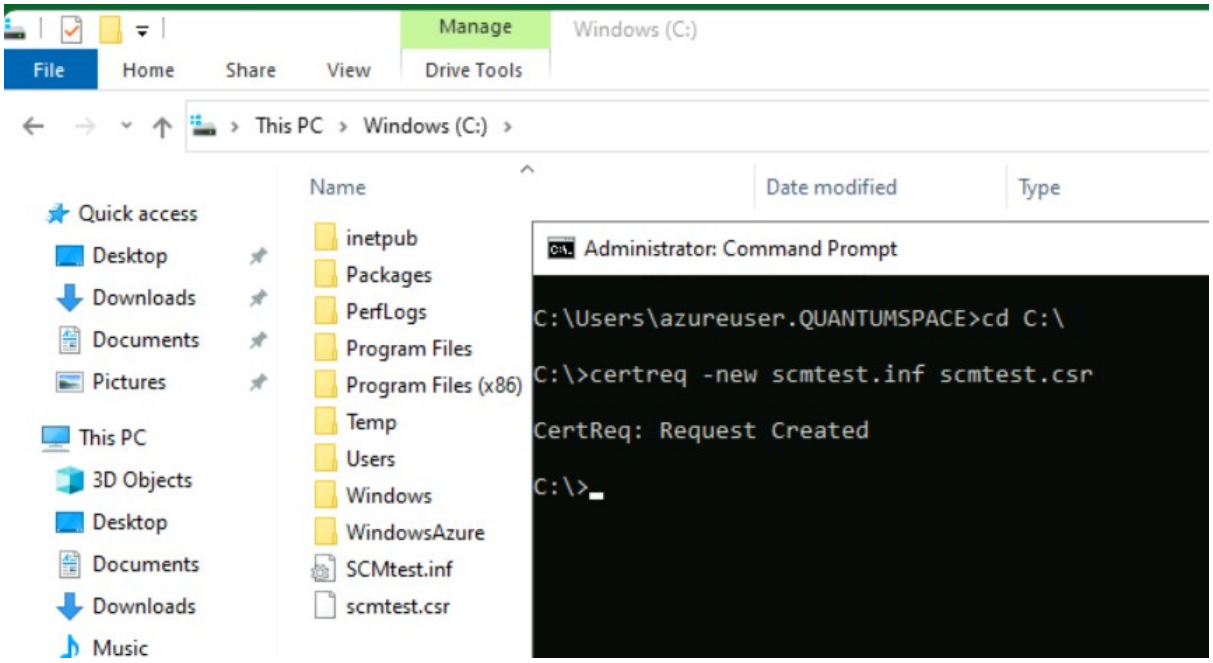


Figure 23. CSR created with certreq

The private key stays on this machine. You need it later for `certreq -accept`.

1b. Submit it through SCM

Sectigo's enrollment form needs a certificate profile and a CSR, and admins can enroll too. Log in to SCM as an MRAO or admin, pick the SSL enrollment form and click Add (+). Menu names in your trial may differ slightly.

1. Select `QS-Test-WebServer`.
2. Paste the CSR content from `scmtest.csr`.
3. Choose a term within what the profile allows.
4. Submit.

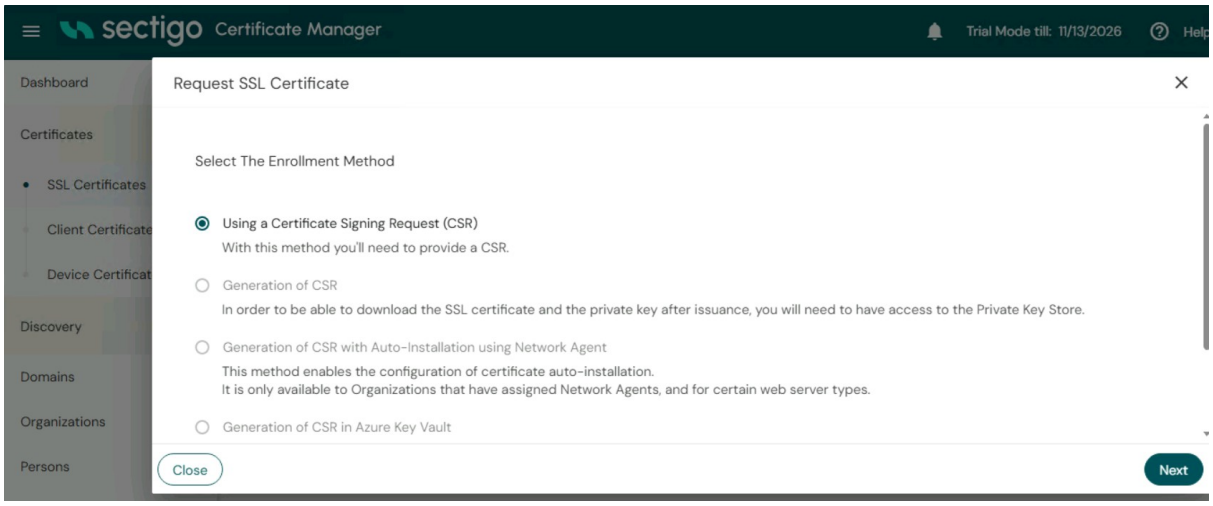


Figure 24. Request SSL Certificate: enrollment method

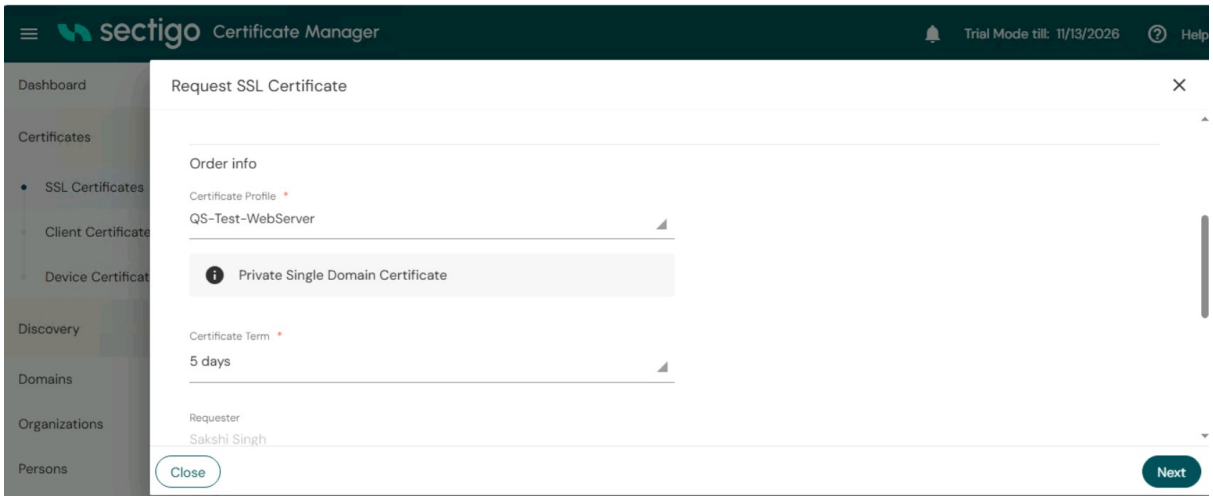


Figure 25. *Request SSL Certificate: profile and term*



Figure 26. *Request SSL Certificate: CSR pasted*

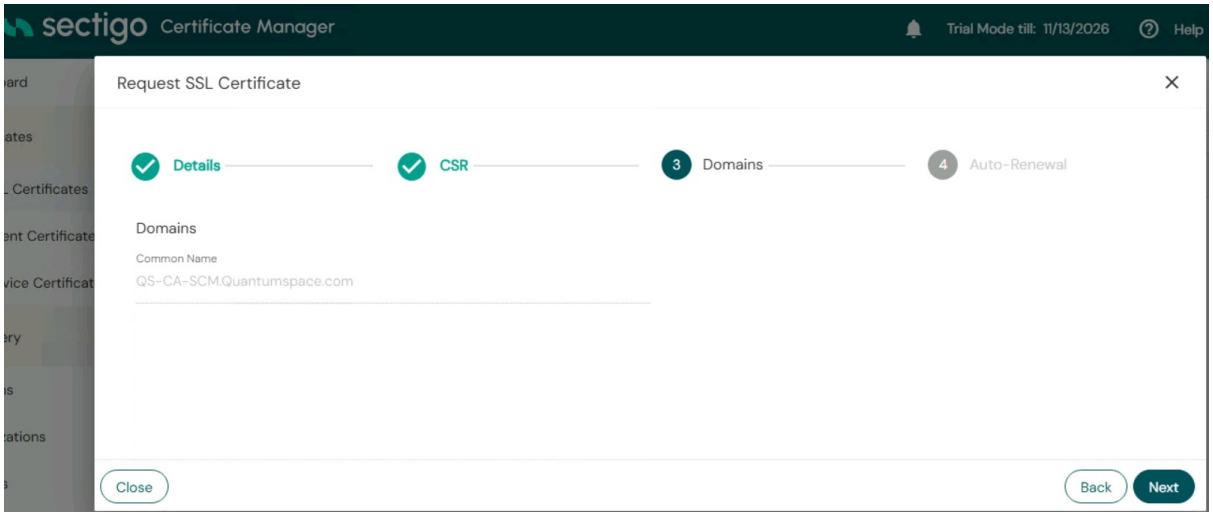


Figure 27. *Request SSL Certificate: domains*

Auto-renewal can be tested at this stage. If renewal needs a new CSR (a new key), change the settings on the certificate profile and on the CA template security as well.

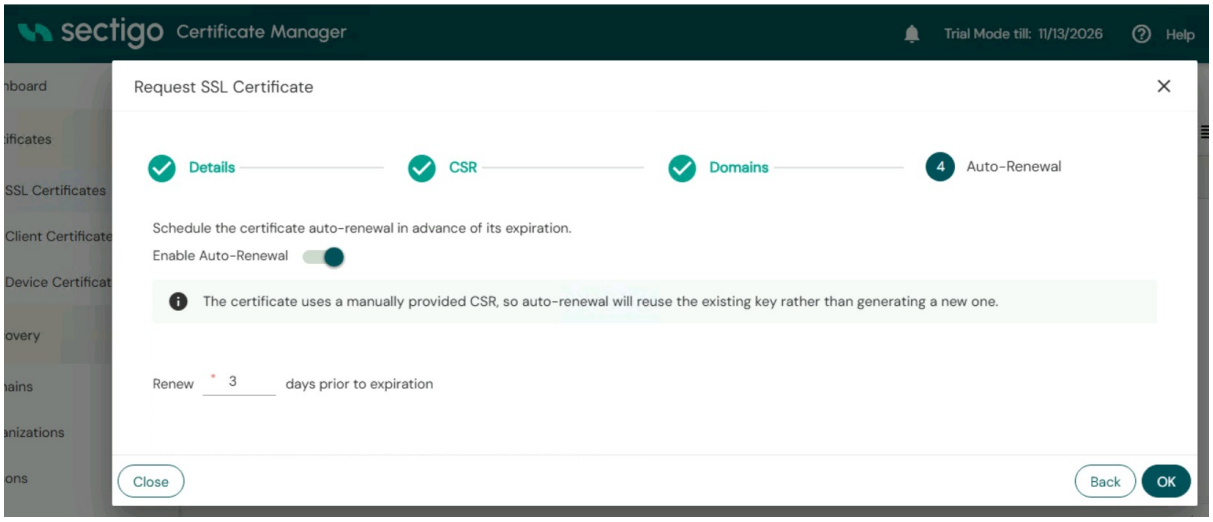


Figure 28. Request SSL Certificate: auto-renewal enabled

For this first test, leave **Enable Auto-Renewal** off.

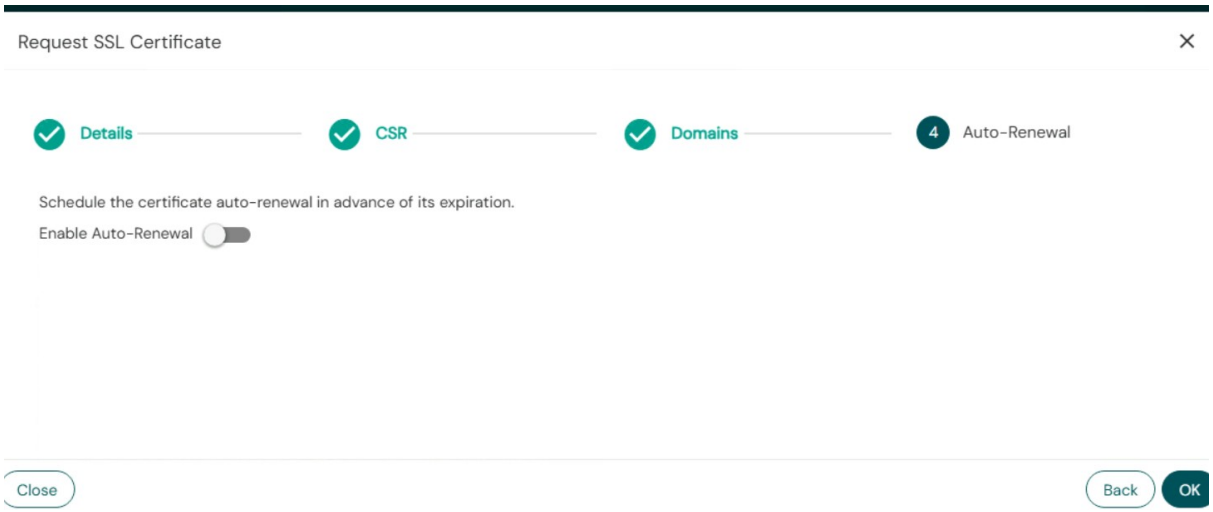


Figure 29. Request SSL Certificate: auto-renewal disabled

1c. Watch the flow

Where	What you should see
SCM	Status changes from Applied to Issued within about a minute or two
CA (`certsrv.msc` → Issued Certificates)	A new row using the `SCM-Test-WebServer` template
Connector logs	Activity in `C:\ProgramData\Sectigo Limited\SectigoCBCS`

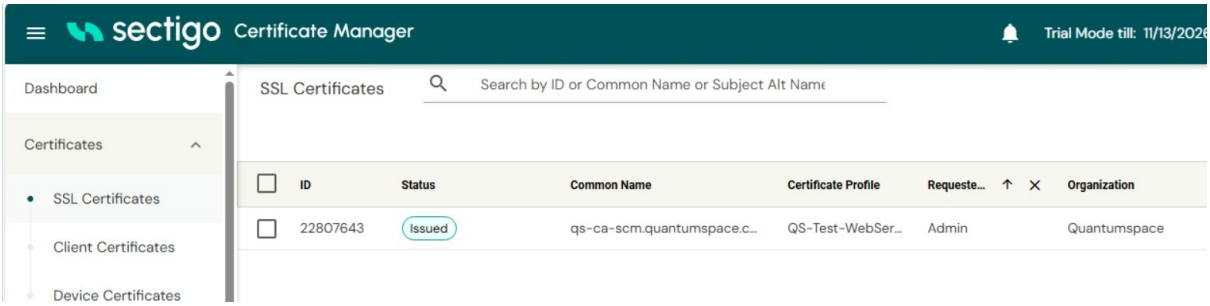


Figure 30. Issued certificate listed in SCM

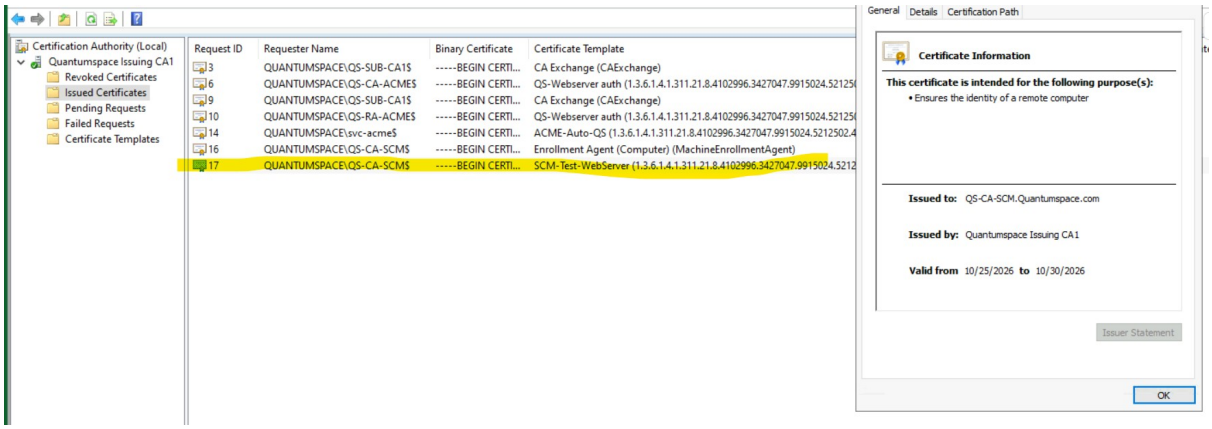


***Figure 31.** Issued certificate on the CA*

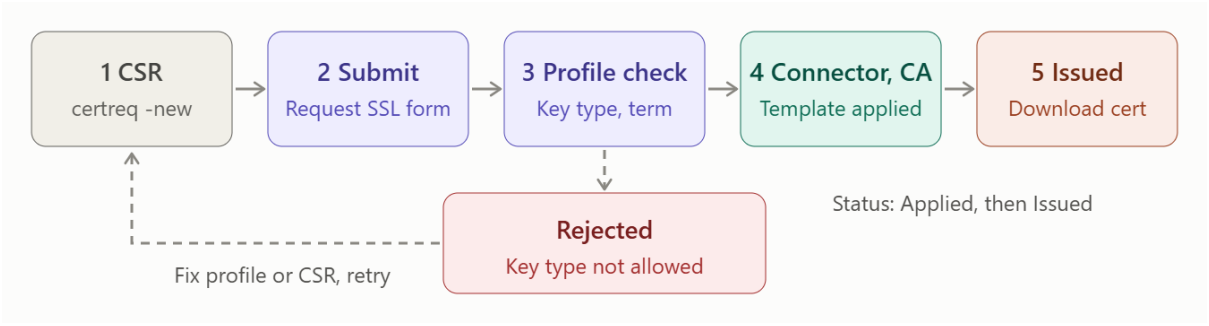


***Figure 32.** Request lifecycle and status in SCM*

SCM also sends an email notification to the designated person's address when the certificate is issued:

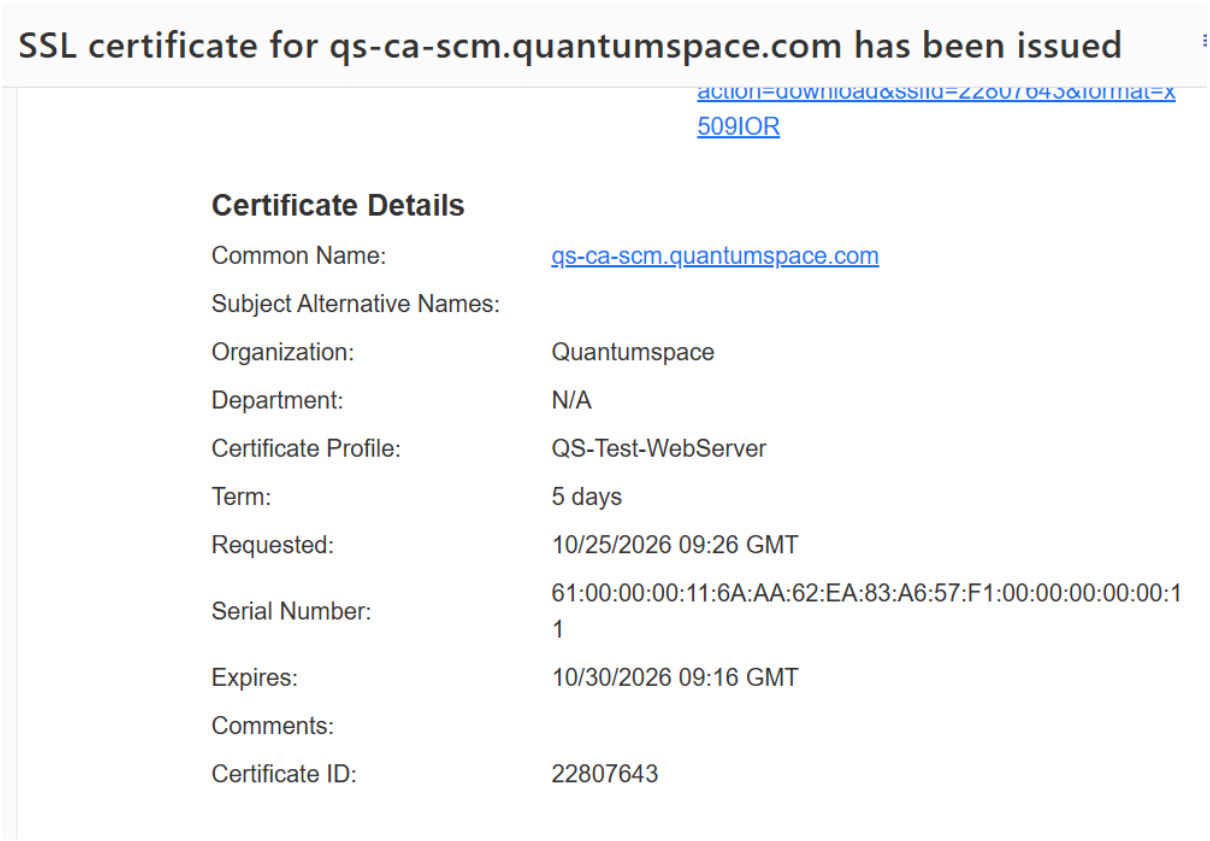


Figure 33. *Issuance notification email*

1d. Retrieve and verify

1. Download the certificate from SCM.
2. Install it where the key lives:

```
certreq -accept issued.cer
```

3. Check the chain and revocation:

```
certutil -verify -urlfetch issued.cer
```

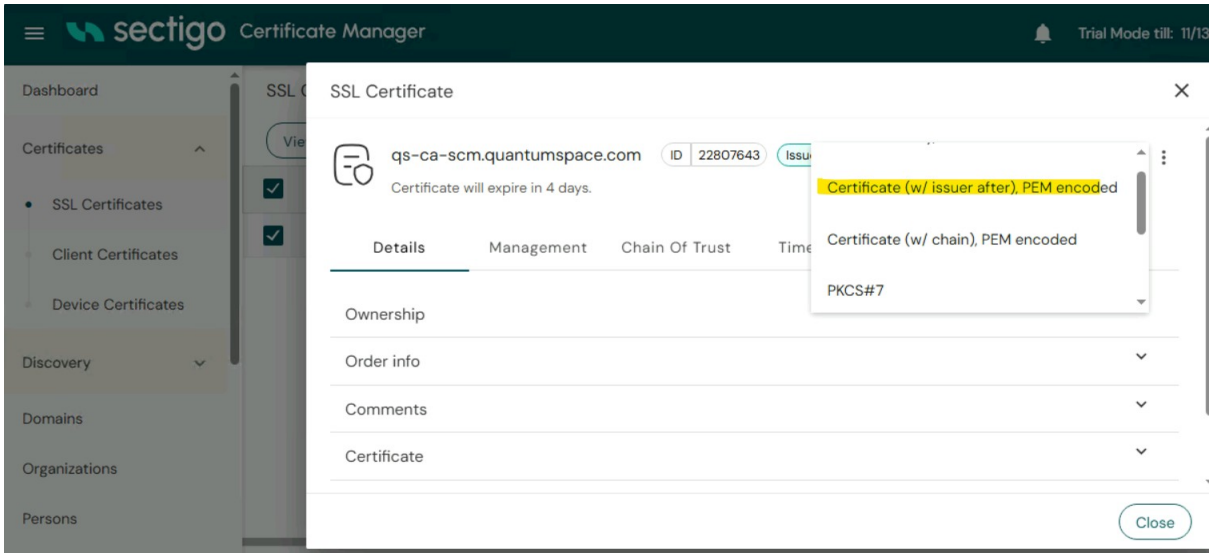


Figure 34. *Certificate download options in SCM*

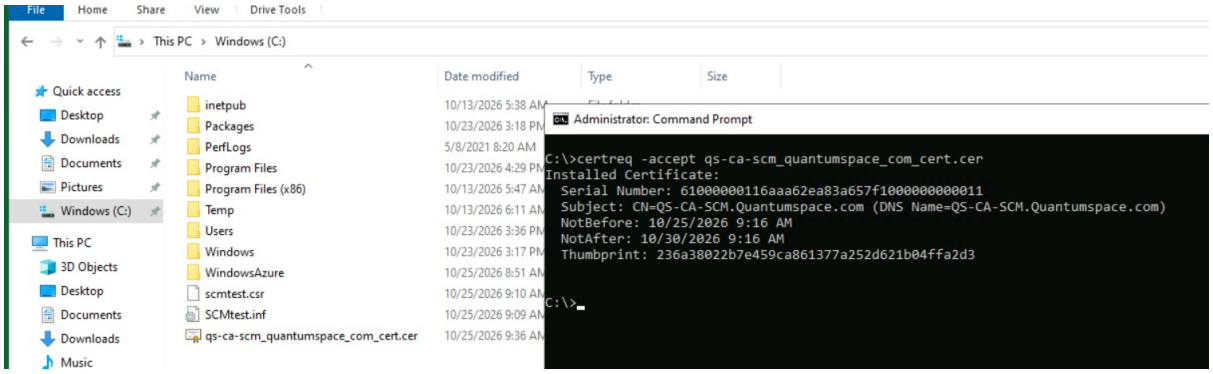


Figure 35. *Certificate accepted with certreq -accept*

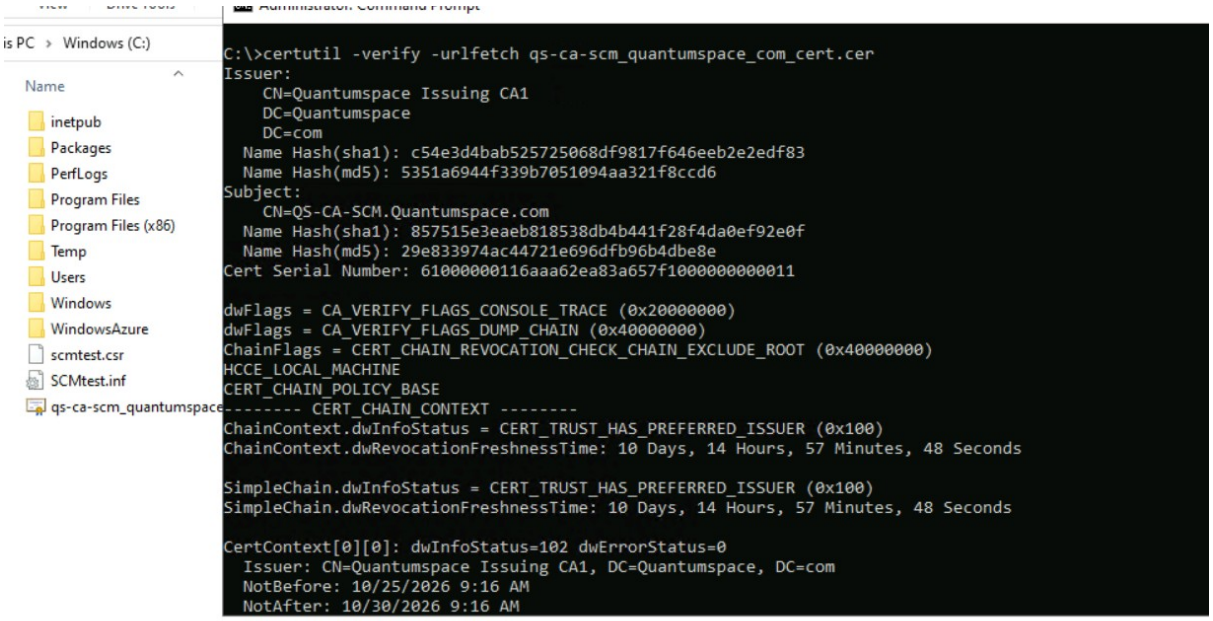


Figure 36. *certutil -verify output: chain*

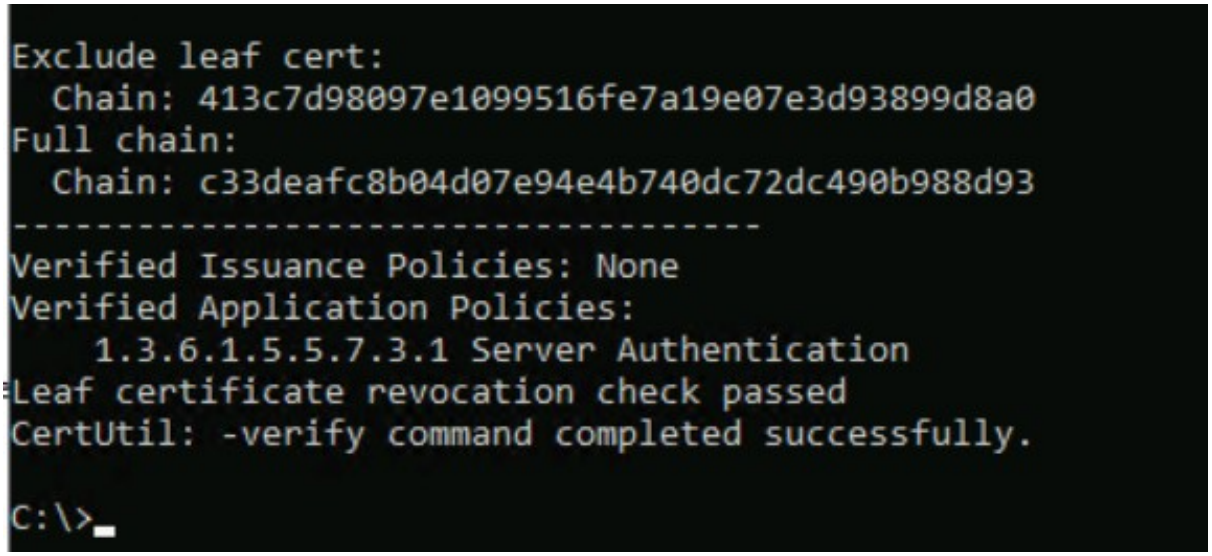


Figure 37. certutil -verify output: revocation check passed

> **Pass criteria:** the chain builds to your root, the SAN matches, and the CRL/AIA checks succeed.

Test 2: Client certificate (optional)

This test is heavier, so do it after Test 1 passes. Sectigo's client certificate flow uses an enrollment form of type **Client certificate self-enrollment**, a person added in SCM, authentication by email confirmation or secret ID, and a download that is usually a PKCS#12 file.

1. Enrollment → Enrollment Forms → create the form and attach `QS-Test-Client`.
2. Add a person under Persons.
3. Open the form URL and authenticate. Use Secret ID if you don't want email to be a variable.
4. Download the PKCS#12 and import it.

Expect the new certificate in the CA's Issued Certificates with the `SCM-Test-Client` template.

Test 3: Revocation

1. In SCM, find the Test 1 certificate and revoke it.

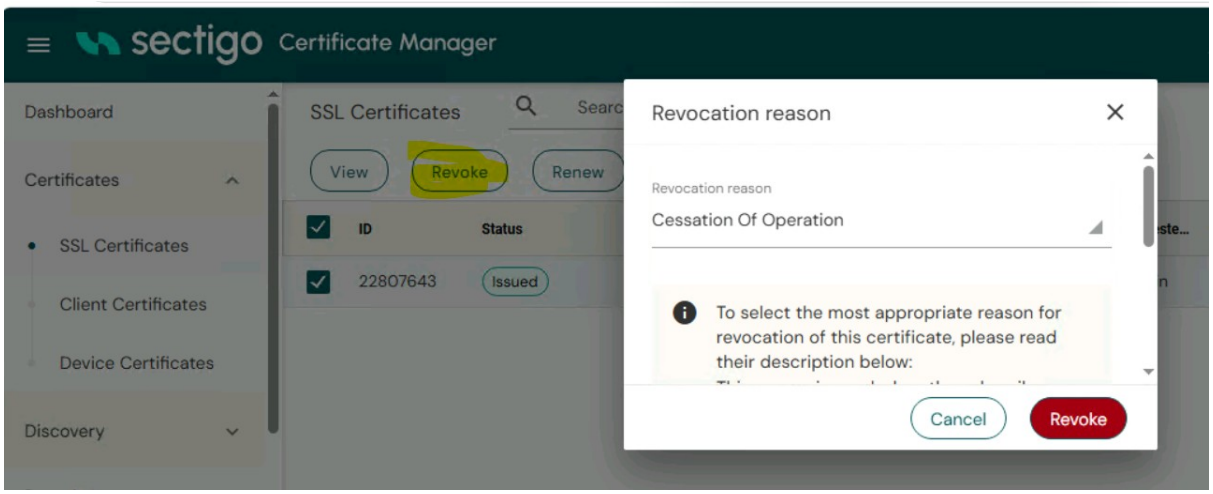


Figure 38. Revoking the certificate in SCM

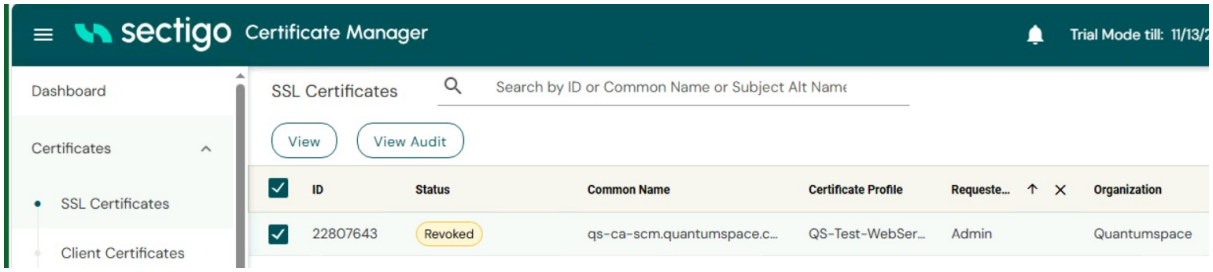


Figure 39. Certificate status Revoked in SCM

1. Confirm it appears under **Revoked Certificates** on the CA.

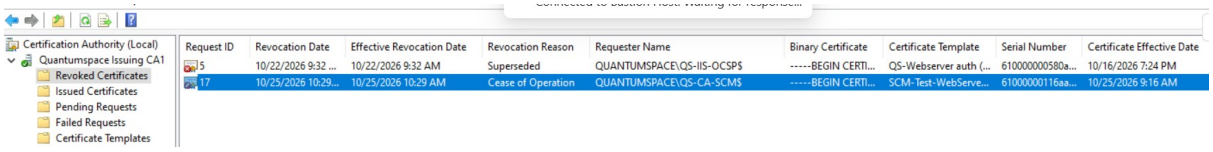


Figure 40. Revoked Certificates on the CA

1. Force a fresh CRL and re-check:

```
certutil -config "QS-SUB-CA1\Quantumspace Issuing CA1" -CRL
certutil -verify -urlfetch issued.cer
```

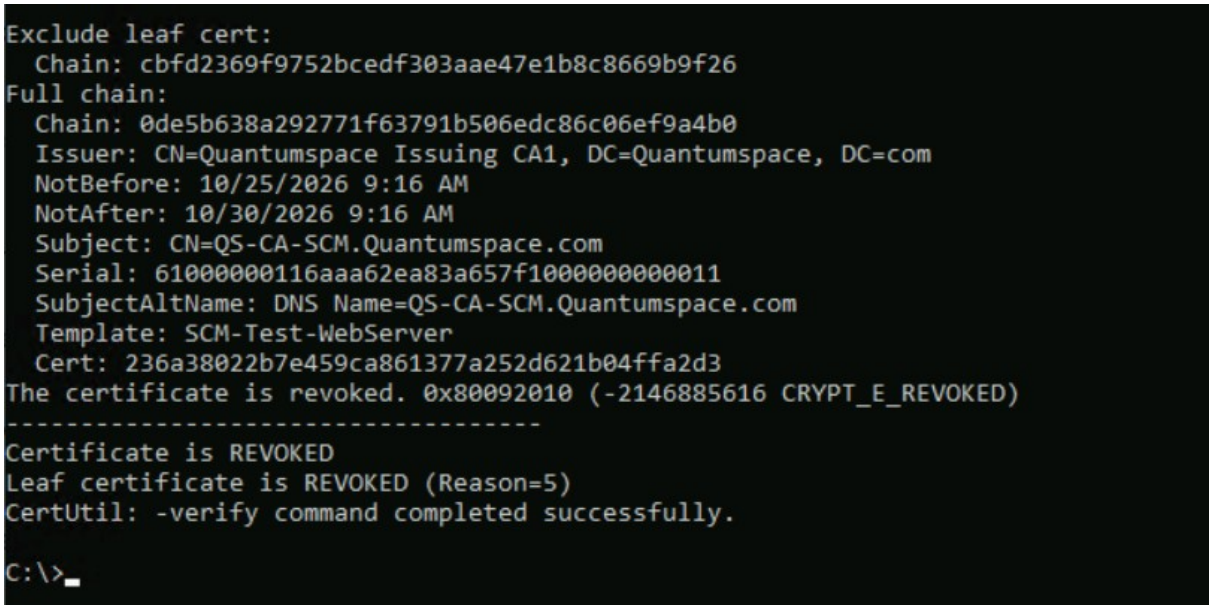


Figure 41. certutil -verify showing the certificate as revoked

Test 4: Renewal

The profile has **Allow Renew** on. Use the renewal option on the certificate in SCM, supply a new CSR, and confirm a second certificate is issued by the CA.

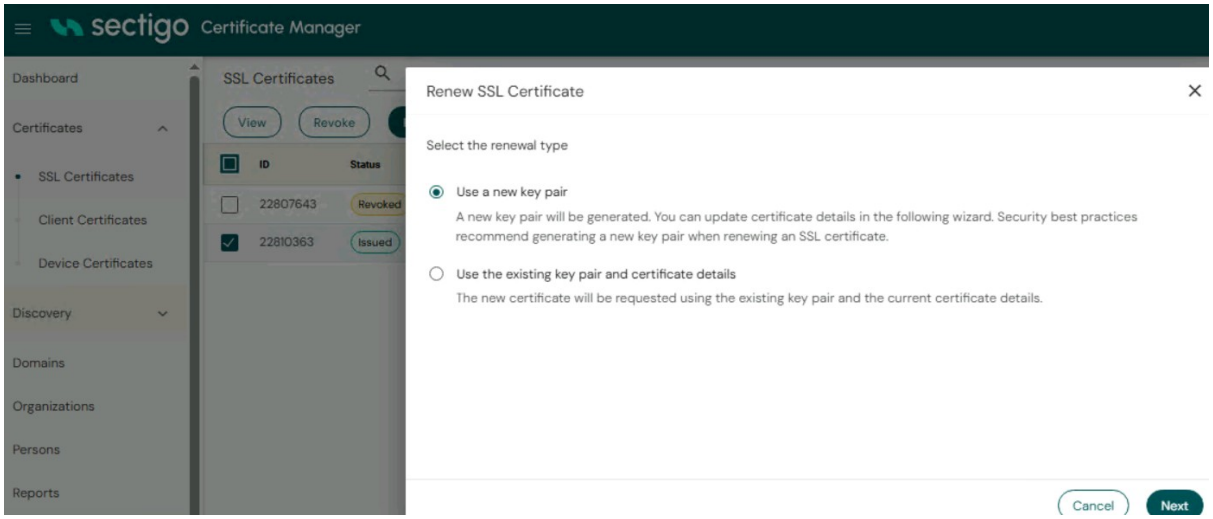


Figure 42. *Renewal options in SCM*

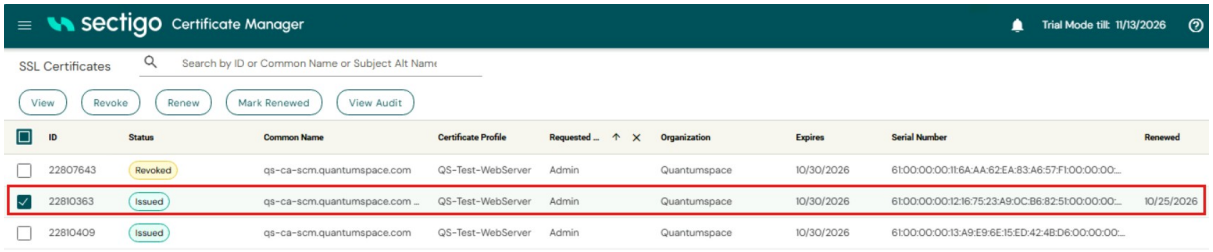


Figure 43. *Original and renewed certificates in SCM*

Troubleshooting

"Failed to initialize RPC connection" on revoke

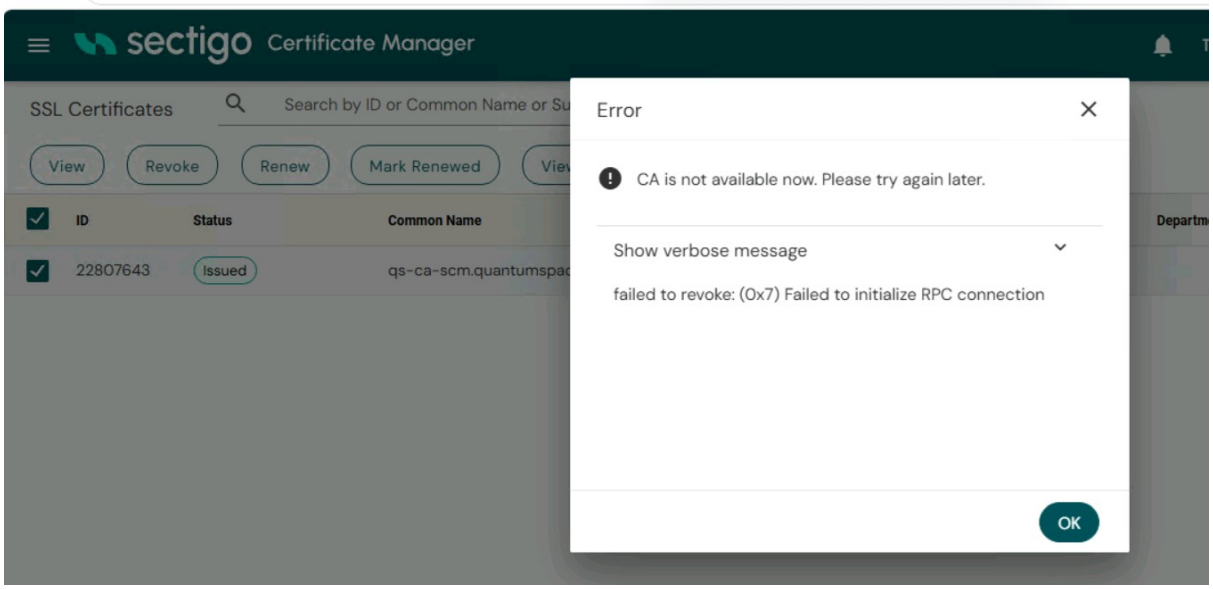


Figure 44. *Error: Failed to initialize RPC connection*

What it tells us. Issuance worked, so basic connectivity and the Enrollment Agent are fine. Revoke uses a different CA interface. Requests go through the certificate request interface, while revoking goes through

the CA admin interface (`ICertAdmin`), which has its own network, DCOM and permission requirements. Sectigo does not document this; the explanation is based on how Microsoft CAs work.

Nothing was revoked. The certificate is still valid.

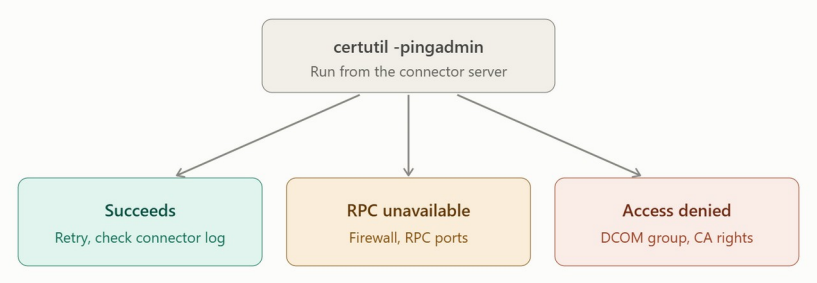


Figure 45. *Decision tree for the -pingadmin result*

1: Test the admin interface directly

From `QS-CA-SCM`, in an elevated prompt:

```
certutil -config "QS-SUB-CA1\Quantumspace Issuing CA1" -pingadmin
```

Result	Likely cause
Succeeds	Not an interface problem. Retry the revoke and read the connector log.
"RPC server is unavailable" (`0x800706BA`)	Network or firewall
"Access is denied" (`0x80070005`)	DCOM or CA permissions

2: If RPC is unavailable

The admin interface uses DCOM, which needs more than port 135.

1. Allow TCP 135 plus the dynamic RPC range (default 49152-65535) from `QS-CA-SCM` to `QS-SUB-CA1`. If your security team wants it tighter, they can pin the DCOM port range instead.
2. On the CA's Windows Firewall, check that **COM+ Network Access (DCOM-In)** and the RPC endpoint mapper rules are enabled.
3. Test the endpoint mapper:

   ```
   Test-NetConnection QS-SUB-CA1 -Port 135
   ```

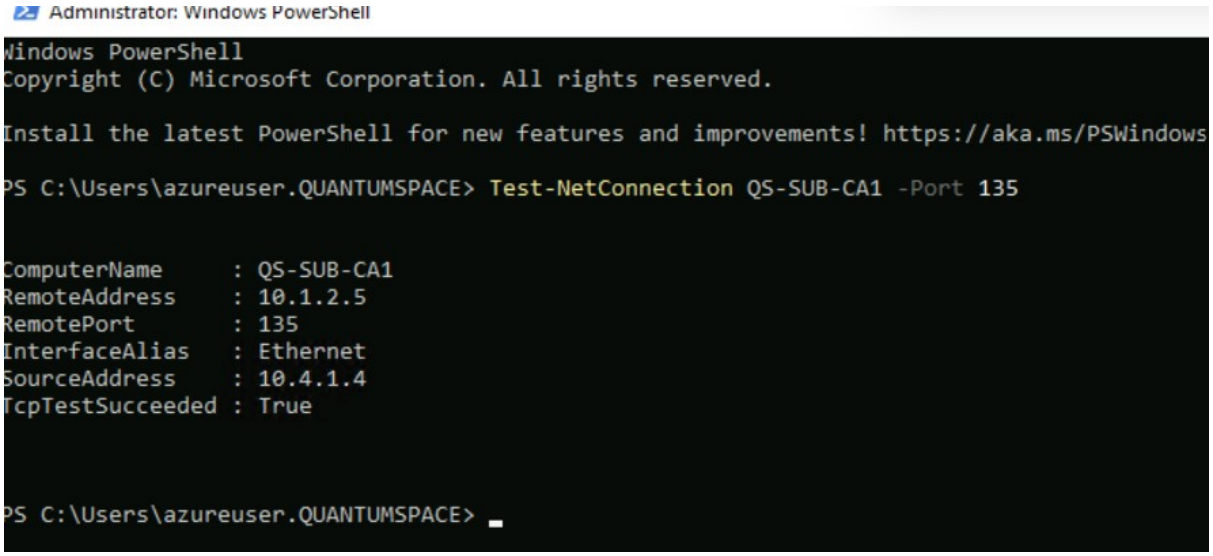


Figure 46. *RPC endpoint mapper test (port 135)*

1. Confirm the CA hasn't been set to block remote admin. On the CA run:

```
certutil -getreg CA\InterfaceFlags
```

Per Microsoft's `InterfaceFlags` documentation, the value `0x10` means remote `ICertAdmin` is disabled. If your value includes that bit, that is the cause. A security team may have set it deliberately, so confirm before changing it.

3: If access is denied

1. Open `certsrv.msc`, right-click the CA, then Properties → Security. Confirm `QS-CA-SCM$` has Allow for both **Manage CA** and **Issue and Manage Certificates**, with no Deny.
2. On the same Properties window, open the **Certificate Managers** tab. If "Restrict certificate managers" is selected, add `QS-CA-SCM$` and allow Everyone as the requester group. If it says "Do not restrict", skip this.
3. Open Computer Management → Local Users and Groups → Groups → **Certificate Service DCOM Access**. Add `QS-CA-SCM$` if it isn't already a member.
4. Restart the CA service:

   ```
   Restart-Service certsvc
   ```

5. In SCM, select the certificate and click Revoke again.

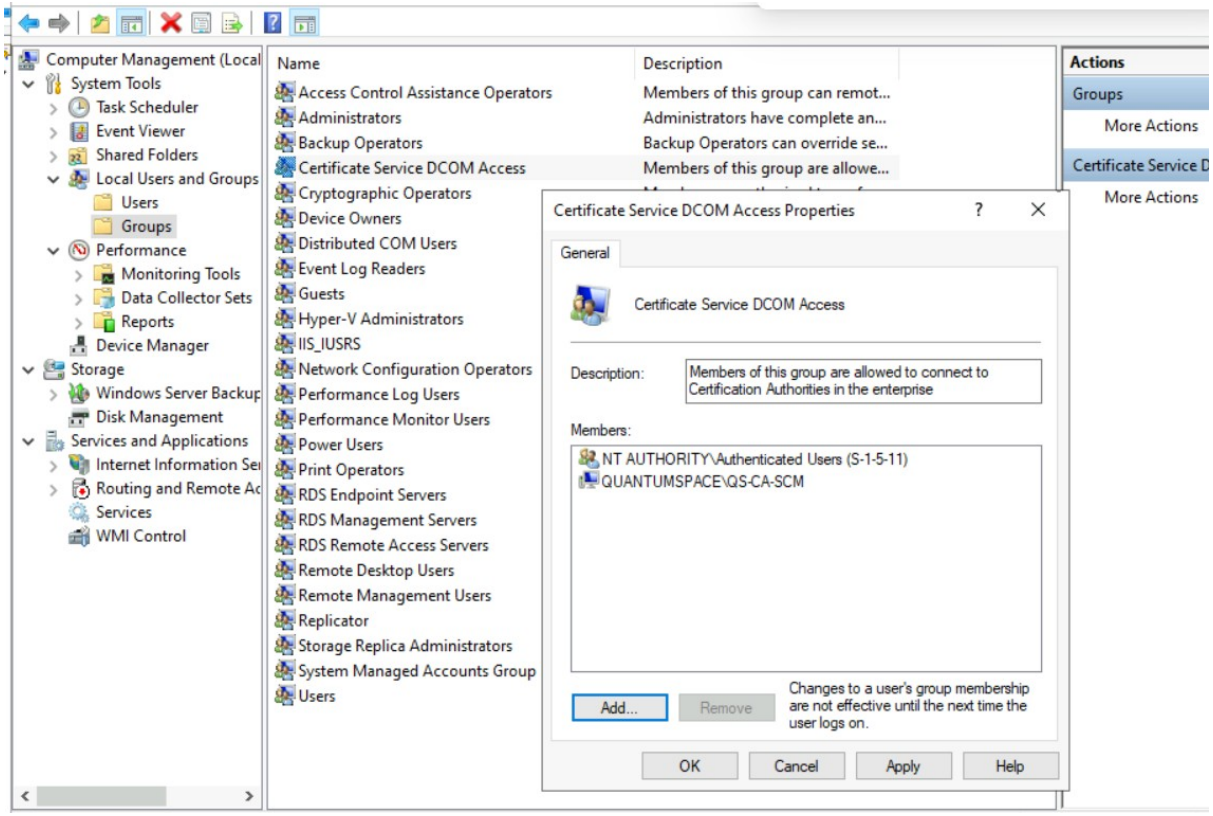


Figure 47. *Certificate Service DCOM Access group on the CA*

If it still fails, read the connector log in `C:\ProgramData\Sectigo Limited\SectigoCBCS`:

```
I [2026-10-25 09:27:41.420+00:00] <2936, 7948> DllMain DLL_THREAD_DETACH
I [2026-10-25 09:28:38.215+00:00] <2936, 3388> DllMain DLL_THREAD_DETACH
I [2026-10-25 09:31:11.097+00:00] <2936, 6660> DllMain DLL_THREAD_ATTACH
I [2026-10-25 09:31:11.113+00:00] <2936, 8164> DllMain DLL_THREAD_ATTACH
I [2026-10-25 09:33:06.247+00:00] <2936, 6660> DllMain DLL_THREAD_DETACH
I [2026-10-25 09:34:13.260+00:00] <2936, 8164> DllMain DLL_THREAD_DETACH
W [2026-10-25 09:47:08.035+00:00] <2936, 3992> ICertAdminD2.RevokeCertificate failed for CA Quantumspace Issuing CA1 certificate 61000000116AAA62EA83A657F1000000000011: 0x80070005 - Access
I [2026-10-25 09:47:08.035+00:00] <2936, 3992> RPC error 0x80070005: trying to re-initialize the connection for context 1
I [2026-10-25 09:47:08.036+00:00] <2936, 7892> DllMain DLL_THREAD_ATTACH
I [2026-10-25 09:47:08.325+00:00] <2936, 3992> RPC error 0x80070005: re-initializing succeeded
W [2026-10-25 09:47:08.352+00:00] <2936, 3992> ICertAdminD2.RevokeCertificate failed for CA Quantumspace Issuing CA1 certificate 61000000116AAA62EA83A657F1000000000011: 0x80070005 - Access
I [2026-10-25 09:47:08.352+00:00] <2936, 3992> RPC error 0x80070005: re-initializing failed
E [2026-10-25 09:47:08.352+00:00] <2936, 3992> Revoke API call failed: 0x7 Failed to initialize RPC connection
W [2026-10-25 09:48:41.036+00:00] <2936, 3992> ICertAdminD2.RevokeCertificate failed for CA Quantumspace Issuing CA1 certificate 61000000116AAA62EA83A657F1000000000011: 0x80070005 - Access
I [2026-10-25 09:48:41.036+00:00] <2936, 3992> RPC error 0x80070005: trying to re-initialize the connection for context 1
I [2026-10-25 09:48:41.212+00:00] <2936, 3992> RPC error 0x80070005: re-initializing succeeded
W [2026-10-25 09:48:41.239+00:00] <2936, 3992> ICertAdminD2.RevokeCertificate failed for CA Quantumspace Issuing CA1 certificate 61000000116AAA62EA83A657F1000000000011: 0x80070005 - Access
I [2026-10-25 09:48:41.239+00:00] <2936, 3992> RPC error 0x80070005: re-initializing failed
E [2026-10-25 09:48:41.239+00:00] <2936, 3992> Revoke API call failed: 0x7 Failed to initialize RPC connection
I [2026-10-25 09:51:17.498+00:00] <2936, 7892> DllMain DLL_THREAD_DETACH
W [2026-10-25 10:00:45.390+00:00] <2936, 3992> ICertAdminD2.RevokeCertificate failed for CA Quantumspace Issuing CA1 certificate 61000000116AAA62EA83A657F1000000000011: 0x80070005 - Access
I [2026-10-25 10:00:45.390+00:00] <2936, 3992> RPC error 0x80070005: trying to re-initialize the connection for context 1
I [2026-10-25 10:00:45.390+00:00] <2936, 1996> DllMain DLL_THREAD_ATTACH
I [2026-10-25 10:00:45.556+00:00] <2936, 3992> RPC error 0x80070005: re-initializing succeeded
W [2026-10-25 10:00:45.583+00:00] <2936, 3992> ICertAdminD2.RevokeCertificate failed for CA Quantumspace Issuing CA1 certificate 61000000116AAA62EA83A657F1000000000011: 0x80070005 - Access
I [2026-10-25 10:00:45.583+00:00] <2936, 3992> RPC error 0x80070005: re-initializing failed
E [2026-10-25 10:00:45.583+00:00] <2936, 3992> Revoke API call failed: 0x7 Failed to initialize RPC connection
```

Figure 48. *Connector log showing the failed revoke*